

O-RAN Test and Integration Focus Group

End-to-End System Testing Framework Specification

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O-RAN Test and Integration Focus Group

End-to-End System Testing Framework Specification

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Revision History

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Date	Revision	Description
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Chapter 1 Scope and Definitions

1.1 Scope

This document is used to define the End-to-End System (E2E) Testing Framework for O-RAN Alliance.

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
 - For a specific reference, subsequent revisions do not apply.
 - For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document.
- [1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”
 - [2] O-RAN Operations and Maintenance Architecture, v03.00, April 2020, O-RAN Alliance
 - [3] O-RAN Architecture Description, v02.00, August 2020, O-RAN Alliance
 - [4] O-RAN Cloud Architecture and Deployment, v02.01, August 2020, O-RAN Alliance
 - [5] O-RAN Cloud Platform Reference Design for Deployment Scenario B, v01.00, April 2020, O-RAN Alliance
 - [6] O-RAN Fronthaul Control, User and Synchronization Plane Specification, v04.00, August 2020, O-RAN Alliance
 - [7] O-RAN Fronthaul Management Plane Specification, v04.00, August 2020, O-RAN Alliance
 - [8] O-RAN Use Cases and Deployment Scenarios White Paper, February 2020, O-RAN Alliance
 - [9] 3GPP TS 28.554: “Management and orchestration; 5G end to end Key Performance Indicators (KPI)”
 - [10] Definition of the Testing Framework for the NGMN 5G Pre-Commercial Network Trials, Version 3, July 2019, NGMN
 - [11] O-RAN Indoor Picocell Hardware Architecture and Requirement (FR1 Only) Specification, v01.00, April 2019, O-RAN Alliance
 - [12] O-RAN Deployment Scenarios and Base Station Classes for White Box Hardware, v01.00, December 2019, O-RAN Alliance

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in this section and the following listed references apply. A term defined in the present document takes precedence over the definition of the same term, if any, in the following listed references.

1. 3GPP TR 21.905 [1]
2. O-RAN Operations and Maintenance Architecture [2]

O-RAN Deployment Blueprint: defined as the set of inputs which is used to describe a specific O-RAN deployment from several aspects. These aspects include the specification of the O-RAN deployment at the System level (e.g. architecture, performance metrics) as well as Subsystem level and the Interfaces between the specified Subsystems. This would then allow definition and documentation of the testing methodology and the test cases using the blueprint specification.

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

CNF	Cloud-Native Network Function
DUT	Device Under Test
E2E	End to End
FHGW	Fronthaul Gateway
FHM	Fronthaul Multiplexer
IUT	Interface Under Test
KPI	Key Performance Indicator
Near-RT RIC	Near-Real Time RAN Intelligent Controller
NMS	Network Management System
Non-RT RIC	Non-Real Time RAN Intelligent Controller
O-Cloud	O-RAN Cloud Platform
O-CU	O-RAN Central Unit
O-CU-CP	O-RAN Central Unit – Control Plane
O-CU-UP	O-RAN Central Unit – User Plane
O-DU	O-RAN Distributed Unit
O-eNB	O-RAN eNB
O-RAN	Open RAN
O-RU	O-RAN Radio Unit
OSC	O-RAN Software Community
PNF	Physical Network Function
RSAC	Requirements and Software Architecture Committee
SLA	Service Level Agreement
SMO	Service Management and Orchestration
SUT	System Under Test
VNF	Virtual Network Function

1.4 Revision Guideline

This Technical Specification has been produced by the O-RAN Alliance.

The contents of the present document are subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of release date and an increase in version number as follows:

Release x.y.z

where:

- x the first digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).
- y the second digit is incremented when editorial only changes have been incorporated in the document.
- z the third digit included only in working versions of the document indicating incremental changes during the editing process.

Chapter 2 Introduction

This document defines the End to End (E2E) testing framework for comprehensive E2E testing and validation of the Open RAN (O-RAN) Systems ensuring robust interoperability and operate with high performance as intended in real-world deployments.

Figure 2-1 shows the focus on the O-RAN System as the System under Test (SUT) in the context of the E2E testing framework specified in this document and in relation to the E2E KPIs defined between the UEs and Services which will be further elaborated in this specification. Refer to [3] and Section 3.1 in this specification for more details on the O-RAN System.

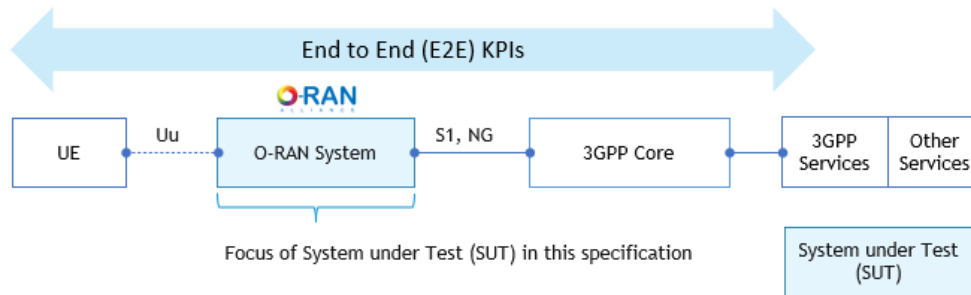


Figure 2-1: Focus on O-RAN as the System under Test (SUT)

Mobile operators will likely plan for many different variants of O-RAN Systems depending on their strategies and use cases which they may be interested to test, deploy and evolve their O-RAN compliant networks over time. The E2E testing framework will therefore need to take into consideration the Operators' O-RAN deployment variants given its influence on the scope of E2E testing.

The concept of O-RAN Deployment Blueprint is introduced in this specification and is used to describe the different deployment variants of O-RAN Systems for E2E system testability purposes. This document provides the definition of the O-RAN Deployment Blueprint template which contains the necessary descriptors to build O-RAN Deployment Blueprints.

A set of typical O-RAN Deployment Blueprints are defined using the most common deployment scenarios abstracted from various surveys which O-RAN Alliance has gathered from global operators. O-RAN Test and Integration Focus Group (TIFG) focuses on the definition of the E2E testing framework based on addressing the needs of these most common and probable deployment scenarios.

Operators and O-RAN Software Community (OSC) have submitted additional detailed inputs on their O-RAN Deployment Blueprints and testing practices respectively, to assist O-RAN Alliance with developing comprehensive E2E testing methodologies and test cases. These inputs are included in this specification for referencing purposes.

Chapter 3 starts with a brief introduction of the O-RAN logical architecture, O-RAN network functions and open interfaces between the O-RAN network functions. These O-RAN network functions and open interfaces are mapped to a list of the most considered use cases driven O-RAN cloud deployment scenarios.

Chapter 4 defines the O-RAN Deployment Blueprint, Template and its associated Profiles which are used to describe the O-RAN Deployment Blueprint. The O-RAN network functions and open interfaces are mapped to the various descriptors of the O-RAN Deployment Blueprint. Detailed definition of the various profiles' templates will be considered in the future releases of this specification.

Chapter 5 details the O-RAN E2E Testing Framework by providing guidance on the Testing Methodology for holistic testing and evaluation of the O-RAN Deployment Blueprints ensuring that these Blueprints can be robustly tested and optimized, ensuring consistent high quality and performance in a deterministic, repeatable and reproducible manner across the entire technology lifecycle.

Chapter 6 provides guidance on the key considerations along with information to help guide testing and validating the O-RAN Deployment Blueprints with the relevant E2E Test Cases and KPIs. Recommendations on the specific E2E Test Cases and KPIs will be considered in the future releases of this specification.

Annex A lists the O-RAN Deployment Blueprints which have been submitted by global operators to O-RAN Alliance with the intention to assist with developing comprehensive E2E testing methodologies and test cases taking into considerations these submitted Blueprints.

Annex B provides an overview on the O-RAN Software Community (OSC) work and its current software releases and testing practices.

Annex C provides an example listing of the test functions and test tools and the applicability these test functions and tools for Subsystem and E2E System Testing.

Annex D references common deployment scenarios abstracted from various surveys which O-RAN Alliance has gathered from global operators and defined by WG6 (Orchestration and Cloudification WG).

The E2E testing framework specified in this document intend to fully leverage the Technical and Test specifications which the O-RAN Alliance Working Groups (WGs) have developed.

Chapter 3 O-RAN Architecture and Deployment Scenarios

This chapter gives an overview of the O-RAN Logical Architecture, O-RAN network functions, open interfaces and a list of the most considered use cases driven O-RAN Deployment Scenarios which show how the O-RAN network functions are mapped to physical implementations in terms of how these network functions are realized as physical, virtual or cloud native network functions, aggregation/dis-aggregation of these network functions and where these network functions can be deployed in the network (e.g. Cell Site, Edge Cloud, Regional Cloud).

The information provided in this Chapter facilitates definition of the O-RAN Deployment Blueprint in Chapter 4. The O-RAN Deployment Blueprint is used to further describe the O-RAN Deployment at the E2E System and Subsystem levels, and the Open Interfaces between the specified Subsystems. This would then allow definition and documentation of the testing methodology and the test cases using the Blueprint specification.

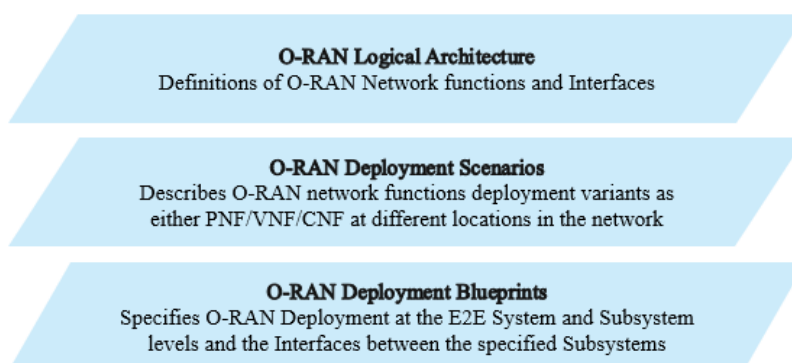


Figure 3-1: O-RAN Logical Architecture, Deployment Scenarios and Deployment Blueprints

3.1 O-RAN Logical Architecture

Figure 3-2 shows the O-RAN logical architecture [3].

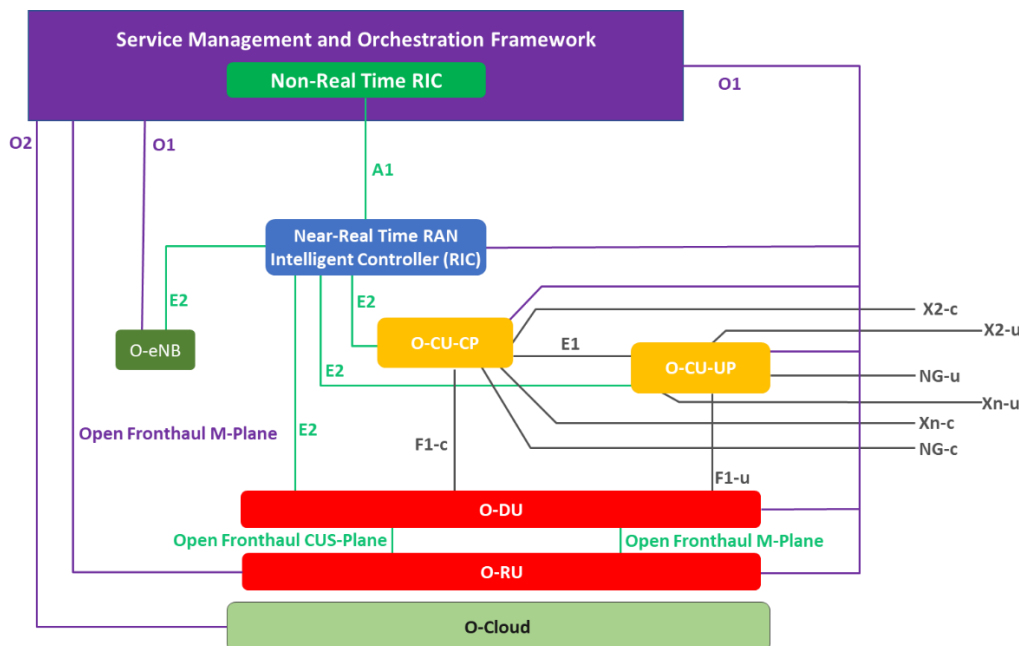


Figure 3-2: Logical Architecture of O-RAN [3]

Refer to O-RAN WG1 “O-RAN Architecture Description” Technical Specification [3] for more details on the definitions for the O-RAN network functions and open interfaces.

These O-RAN network functions and open interfaces are specified by different WGs in O-RAN Alliance.

Table 3-1 shows the listing of the O-RAN network functions and the WGs which are responsible for specifications of the various aspects of these O-RAN network functions.

Table 3-1: O-RAN network functions and WGs responsible for specification

O-RAN network function	O-RAN WG(s)	O-RAN WG focus
O-RU	WG1	O1 management services
	WG4	Open fronthaul CUS and M plane; Management Plane O1 alignment work is in progress
	WG7	Whitebox reference design
O-DU	WG1	O1 management services
	WG3	E2 node
	WG4	Open fronthaul CUS and M plane; Management Plane
	WG5	C and U-Plane profiles for F1 O1 information and data models
	WG6	O-Cloud APIs (including AAL APIs)
	WG7	Whitebox reference design
	WG8	Reference Stack: O-DU Software Architecture and APIs
O-CU-CP	WG1	O1 management services
	WG3	E2 node
	WG5	C-Plane profiles for X2, Xn, F1, E1 O1 information and data models
	WG6	O-Cloud APIs (including AAL APIs)
	WG7	Whitebox reference design
	WG8	Reference Stack: O-CU-CP Software Architecture and APIs
O-CU-UP	WG1	O1 management services
	WG3	E2 node
	WG5	U-Plane profiles for X2, Xn, F1 C-Plane profile for E1 O1 information and data models
	WG6	O-Cloud APIs (including AAL APIs)
	WG7	Whitebox reference design
	WG8	Reference Stack: O-CU-UP Software Architecture and APIs
O-eNB	WG1	O1 management services
	WG3	E2 node
	WG4	Open fronthaul CUS and M plane; Management Plane includes O1 alignment
	WG5	C and U-Plane profiles for X2, Xn O1 information and data models
	WG6	O-Cloud APIs (including AAL APIs)
	WG7	Whitebox reference design
O-Cloud	WG6	O2 O-Cloud APIs (including AAL APIs)
Near-Real Time RIC	WG1	O1 management services

O-RAN network function	O-RAN WG(s)	O-RAN WG focus
	WG3	E2 xApp Open APIs O1 information and data models
	WG6	O-Cloud APIs (including AAL APIs)
SMO (including Non-Real Time RIC function)	WG1	O1 interface and common information/data models
	WG2	A1

Table 3-2 shows the listing of the O-RAN Open Interfaces and the WGs which are responsible for the specifications for these O-RAN open interfaces.

Table 3-2: O-RAN Open Interfaces and WGs responsible for specification

O-RAN interfaces	O-RAN WG(s)	O-RAN WG focus
A1	WG2	Specifies Control plane protocols for the logical interface between Non-RT-RIC function and near-RT-RIC function.
E1	WG5	3GPP defined logical interface between gNB-CU-CP and gNB-CU-UP. O-RAN WG5 reuses and adopts these principles and protocol stack defined by 3GPP for the O-CU-CP and the O-CU-UP functions, as well as for the definition of interoperability profile specifications.
E2	WG3	Specifies Control plane protocols for the logical interface between near-RT-RIC function and E2 Nodes for E2 functions.
F1-C	WG5	3GPP defined logical interface between gNB-CU-CP and gNB-DU. O-RAN WG5 reuses and adopts these principles and protocol stack defined by 3GPP for the O-CU-CP and the O-DU functions, as well as for the definition of interoperability profile specifications.
F1-U	WG5	3GPP defined logical interface between gNB-CU-UP and gNB-DU. O-RAN WG5 reuses and adopts these principles and protocol stack defined by 3GPP for the O-CU-UP and the O-DU functions, as well as for the definition of interoperability profile specifications.
O1	WG1	Specifies Management plane interface between SMO and O-RAN Managed Element. Noting that the Information and Data models for the O-RAN Managed Elements are specified by their respective WGs shown in Table 3-1.
O2	WG6	Specifies Management plane interface between SMO and O-Cloud.
Open Fronthaul (OFH)	WG4	Specifies Open fronthaul interface between O-DU and O-RU includes Control, User, Synchronization and Management Planes. Management Planes include both Hierarchical and Hybrid modes. Specifies Open fronthaul M-Plane between the O-RU and SMO for the O-RU configured in Hybrid mode.
X2-C	WG5	3GPP defined logical interface for transmitting control plane information between eNBs or between eNB and en-gNB in EN-DC deployment scenario. O-RAN WG5 reuses and adopts the principles and protocol stack defined by 3GPP for the definition of interoperability profile specifications.
X2-U	WG5	3GPP defined logical interface for transmitting user plane information between eNBs or between eNB and en-gNB in EN-DC deployment scenario. O-RAN WG5 reuses and adopts the principles and protocol stack defined by 3GPP for the definition of interoperability profile specifications.

O-RAN interfaces	O-RAN WG(s)	O-RAN WG focus
Xn-C	WG5	3GPP defined logical interface for transmitting control plane information between gNBs, ng-eNBs or between eNB and ng-gNB. O-RAN WG5 reuses and adopts the principles and protocol stack defined by 3GPP for the definition of interoperability profile specifications.
Xn-U	WG5	3GPP defined logical interface for transmitting user plane information between gNBs, ng-eNBs or between eNB and ng-gNB. O-RAN WG5 reuses and adopts the principles and protocol stack defined by 3GPP for the definition of interoperability profile specifications.

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3.2 O-RAN Cloud Deployment Scenarios

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This section gives a short introduction to each of the O-RAN Cloud Deployment Scenarios abstracted from inputs O-RAN Alliance have received from global operators which have gathered the most interest and are of either current or relatively near-term interest that can be supported by the O-RAN specifications.

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The key differences between these Cloud Deployment Scenarios are highlighted given that these are important factors to be taken into considerations when designing the O-RAN Deployment Blueprint template and the E2E testing methodologies in the next chapters of this specification.

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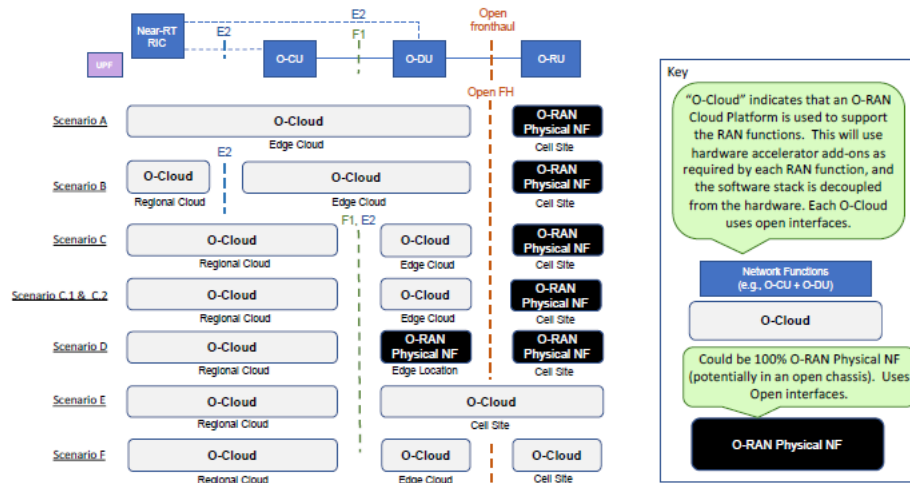
Refer to O-RAN WG6 “Cloud Architecture and Deployment Scenarios for O-RAN Virtualized RAN” Technical Report [4] for more details.

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Figure 3-3 shows a high-level comparison of O-RAN Cloud Deployment Scenarios for NR Standalone (SA) operation [4].

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Figure 3-3: High Level Comparison of O-RAN Cloud Deployment Scenarios [4]

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Few notes for clarification purposes on the O-RAN network functions and interfaces shown in Figure 3-3 and for the remaining of this section.

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1. The O-CU and vO-CU referred in [4] should be considered as O-CU CP/UP and vO-CU CP/UP, respectively.
2. O-RAN deployments with all Physical Network Function (PNF) based implementation may exist but are not shown in this figure as this figure shows cloud deployment scenarios with at least one O-RAN network function implemented on the O-Cloud.
3. O-RAN deployments with PNF instead of VNF/CNF based implementation may exist but are not shown in this figure such as Scenario E with O-DU and O-RU implemented as PNFs (instead of VNF/CNF) at the Cell Site.
4. Not all the O-RAN network functions and interfaces defined by O-RAN Alliance are shown in Figure 3-3. For example, the Non-Real Time RIC, SMO, O-CU-CP, O-CU-UP, O-eNB are not shown in this figure.

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NR Standalone (SA)

1. **(SA) Scenario A** shows the near-RT RIC, O-CU, and O-DU network functions virtualized and co-located on the same Edge Cloud platform. The O-RU deployed at the Cell site is not virtualized.
2. **(SA) Scenario B** is similar to (SA) Scenario A, the only difference is that the Near-Real Time RIC in (SA) Scenario B is deployed on the Cloud Platform at the Regional cloud (as compared to the Edge cloud in (SA) Scenario A).
3. **(SA) Scenario C** shows near-RT RIC, O-CU are virtualized on the same Regional Cloud platform and the O-DU virtualized on an Edge Cloud platform
4. **(SA) Scenario C.1** shows a deployment variant of (SA) Scenario C adding the flexibility to be able to deploy parts of the O-CU-UP in the Edge Cloud platform in order to better support use cases such as differentiated performance requirements of the traffic types and network slices.
5. **(SA) Scenario C.2** shows a deployment variant of (SA) Scenario C adding the flexibility to be able to deploy additional O-DUs and O-CUs in the Edge Cloud platform to better support use cases such as differentiated performance requirements of the traffic types, network slices and RAN sharing.
6. **(SA) Scenario D** shows a deployment variant of (SA) Scenario C, but in this case the O-DU functionality is supported by an O-RAN PNF rather than an O-Cloud.
7. **(SA) Scenario E** shows a future state scenario assuming that the O-RU can be successfully virtualized and deployed with the O-DU in the same O-Cloud platform located at the Cell site. Near-RT RIC and O-CU are virtualized on the same Regional Cloud platform.
8. **(SA) Scenario F** shows a future state scenario assuming that the O-RU can be successfully virtualized and deployed in the O-Cloud platform located at the Cell Site. The near-RT RIC, O-CU are virtualized on the same Regional Cloud platform and the O-DU virtualized on an Edge Cloud platform.

NR Non-Standalone (NSA)

The following listed (NSA) Cloud Deployment scenarios are similar to the (SA) Cloud Deployment scenarios above listed with the LTE network functions (O-CU, O-DU) co-located with the NR network functions (O-CU, O-DU).

1. **(NSA) Scenario A** is similar to (SA) Scenario A with the addition of LTE network functions (O-CU, O-DU) virtualized and co-located on the same Edge Cloud platform as the NR network functions (O-CU, O-DU).
2. **(NSA) Scenario B** is similar to (SA) Scenario B with the addition of LTE network functions (O-CU, O-DU) virtualized and co-located on the same Edge Cloud platform as the NR network functions (O-CU, O-DU).
3. **(NSA) Scenarios C, C.2** are similar to (SA) Scenario C, C.2 with the addition of LTE network function(s) (O-CU) virtualized and co-located on the same Regional Cloud platform as the NR network function(s) (O-CU). The LTE network function(s) (O-DU) virtualized and co-located on the same Edge Cloud platform as the NR network function(s) (O-DU).
4. **(NSA) Scenario D** is similar to (SA) Scenario D with the addition of LTE network function(s) (O-CU) virtualized and co-located on the same Edge Cloud platform as the NR network function(s) (O-CU). The LTE network function(s) (O-DU) is non-virtualized and co-located at the same Edge location as the NR network function(s) (O-DU).

Figures 3-4 and 3-5 show the Cloud Deployment Scenarios B from [4] for NR Standalone (SA) and Non-Standalone (NSA) operations, respectively. Refer to Annex D in this document for additional Cloud Deployment Scenarios figures.

Refer to [4] for the definitions and more details on the GW (Opt) node, RU, vO-CU, vO-DU shown in these figures. Refer to [5] for the Cloud Platform Reference Design for Deployment Scenario B.

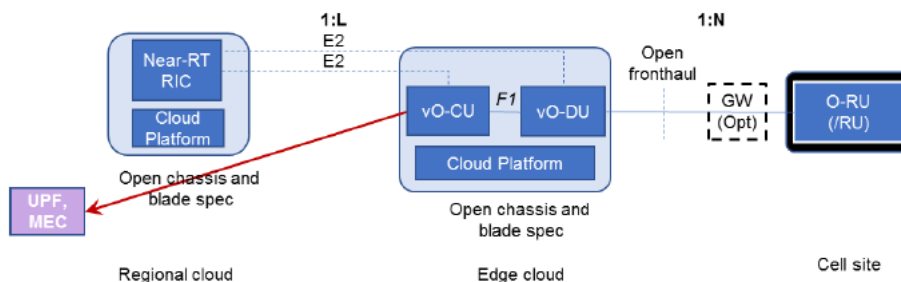


Figure 3-4: Cloud Deployment Scenario B for SA operation [4]

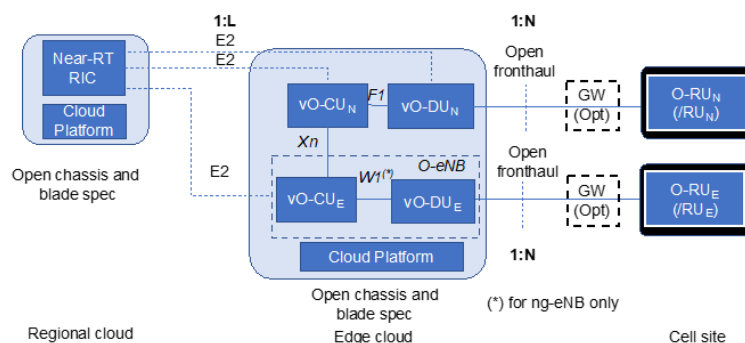


Figure 3-5: Cloud Deployment Scenario B for NSA operation [4]

Noting that all the Cloud Deployment Scenarios listed in [4] use the same set of O-RAN network functions and Open interfaces, while the key differences are with the

1. Use Cases and Services Requirements (e.g. network slicing, bandwidth, latency, reliability etc.).
2. Network design considerations (e.g. transport capabilities such capacity, latency between Cell site, Edge and Regional cloud, operating spectrum bands etc.).
3. NR deployment modes – SA, NSA.
4. Network functions – Physical, Virtual, Cloudified implementation (e.g. taking into considerations O-Cloud solution availability with certain timeframes in mind).
5. Co-locations of the various O-RAN network functions (e.g. vO-CU, vO-DU at the Edge Cloud in Scenario B).
6. Co-located O-RAN network functions may be deployed as Integrated or Separated nodes.
7. O-RAN network functions shown are logical functions which in deployment scenarios where more than a single O-RAN network functions are co-located, can be either implemented as an integrated Subsystem (combined vO-CU and vO-DU at the Edge Cloud in Scenario B) or independent Subsystems (separate vO-CU and vO-DU at the Edge Cloud in Scenario B). Refer to Chapter 4 for the definition of the Subsystem.
8. Locations of the O-RAN network functions at the Cell site, Edge cloud, Regional cloud.

Chapter 4 O-RAN Deployment Blueprints

4.1 Introduction

This chapter provides the high-level definition of the O-RAN Deployment Blueprint template which contains the necessary descriptors to build the underlying O-RAN Deployment Blueprints.

The O-RAN Deployment Blueprint is defined as the set of inputs used to describe a specific O-RAN deployment from several aspects. These aspects include the specification of the O-RAN deployment at the System level (e.g. architecture, performance metrics) as well as Subsystem level and the Open Interfaces between the specified Subsystems. This would then allow definition and documentation of the testing methodology and the test cases using the blueprint specification.

Each of the O-RAN Deployment Blueprints can be described with

1. A single System Profile
2. Two or more Subsystem Instance Profiles
3. One or more Subsystem Pair Interoperability Profile(s)

Figure 4-1 shows the relationship between Deployment Blueprint and its associated set of Profiles.

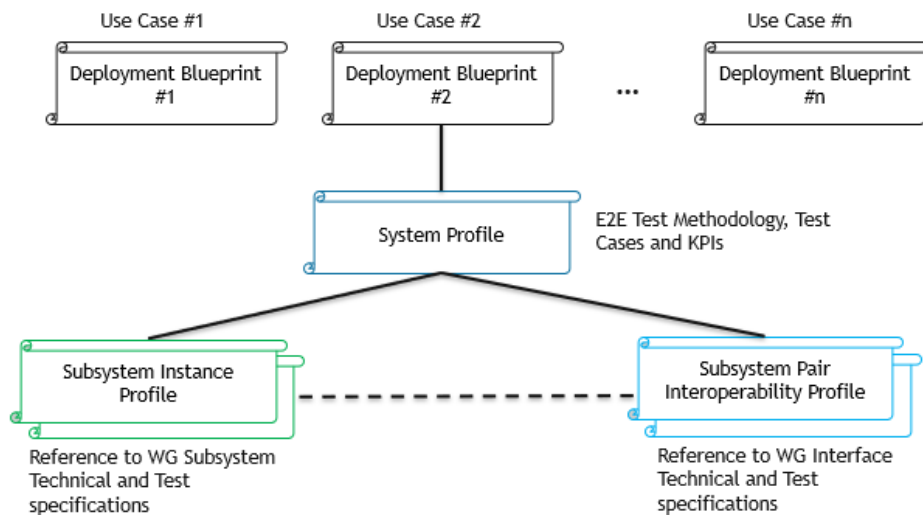


Figure 4-1: Deployment Blueprint and the associated Profiles

4.2 System Profiles

The purpose of the System Profile is to describe the O-RAN Deployment Blueprint from the E2E system perspective.

A **System Profile** consists of **System Architecture information** and a set of **System Profile parameters**.

Figure 4-2 shows the scope of the System Profile in relation to the O-RAN Deployment Blueprint.

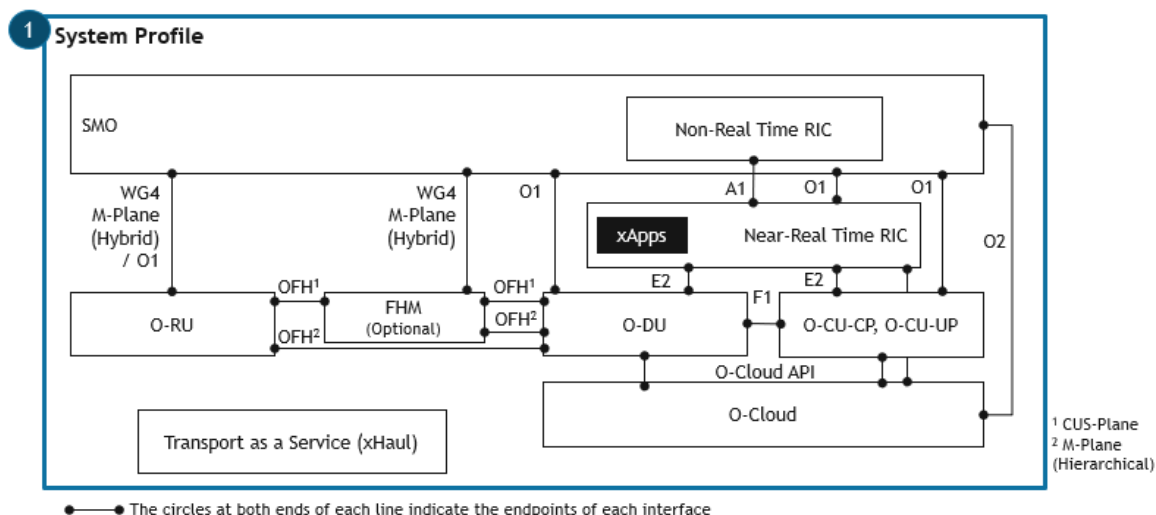


Figure 4-2: Scope of the System Profile in relation to the Deployment Blueprint

Note that the intent of Figure 4-2 is used to show the scope of the System Profile in relation to the Deployment Blueprint. It is not intending to show the actual O-RAN network functions and open interfaces which must be included in the System Profile for all the O-RAN Deployment Blueprints.

OFH indicated in Figure 4-2 includes WG4 CUS and M plane implementation and for M-Plane both the Hierarchical and Hybrid models. Refer to [6][7] for more details.

System Architecture Information shall include an Architecture diagram which is used to describe the Network and Network Slices Topology which consists of the

1. Subsystems involved
2. Type of implementation of each Subsystem (e.g. as a Physical (PNF), Virtual (VNF) or Cloud-Native Network Function (CNF))
3. Location where each of the Subsystems are expected to be deployed (e.g. Cell site, Edge site, Regional site, Core/Central site as appropriate)
4. Interconnections and Cardinality between the Subsystems

The intent of this specification is to make references as much as possible to the list of Deployment Scenarios documented in O-RAN WG6 “Cloud Architecture and Deployment Scenarios for O-RAN Virtualized RAN” Technical Report [4].

Figure 3-4 shows an example of the System Architecture Diagram for Cloud Scenario B referenced from [4].

It should be noted that there can be O-RAN Deployment Blueprints which may be designed for deployment scenarios currently not specified in [4]. Examples can be an all PNF based O-RAN deployment, Non-Standalone (NSA) O-RAN deployment, different xApps implementation among others.

In addition to specifying the System Architecture diagram and information, it would be useful to specify the following listed **System Profiles parameters** which are used to describe the System level requirements for the Blueprints.

1. Services Requirements (e.g. eMBB, URLLC, mIoT, combinations of these)
2. Use Cases (e.g. 3GPP use cases such as VoLTE/VoNR, O-RAN RIC-enabled use cases such as Traffic Steering, QoE Optimization, Massive MIMO Beamforming Optimization, RAN Sharing among others – refer to [8] for more details)
3. Use Cases E2E Performance requirements of the Application, Services (e.g. data rate, latency, reliability, availability, deterministic networking including transmission latency etc.)
4. Indoor or Outdoor deployment
5. Radio Access Technologies (RAT) Types (e.g. 4G LTE/5G NR)
6. 5G NR Deployment Options and Evolution (options of Standalone SA, Non-Standalone NSA)
7. Spectrum Bands and Duplex Modes for 4G LTE/5G NR
8. Cell Types (e.g. Macro, Micro, Pico, others)

9. Network and Network Slices Topology needing to include Cardinality, Network functions (xNFs), Transport (T-plane), Timing and Sync (S-Plane), Management (M Plane, MANO, SMO)
10. E2E Network Performance requirements of the RAN Network Functions, Slices and Management KPIs

4.3 Subsystem Instance Profiles

Each of the Subsystem in the Deployment Blueprint can be described with its **Subsystem Instance Profile**.

Figure 4-3 shows the scope of the Subsystem Instance Profile in relation to the O-RAN Deployment Blueprint.

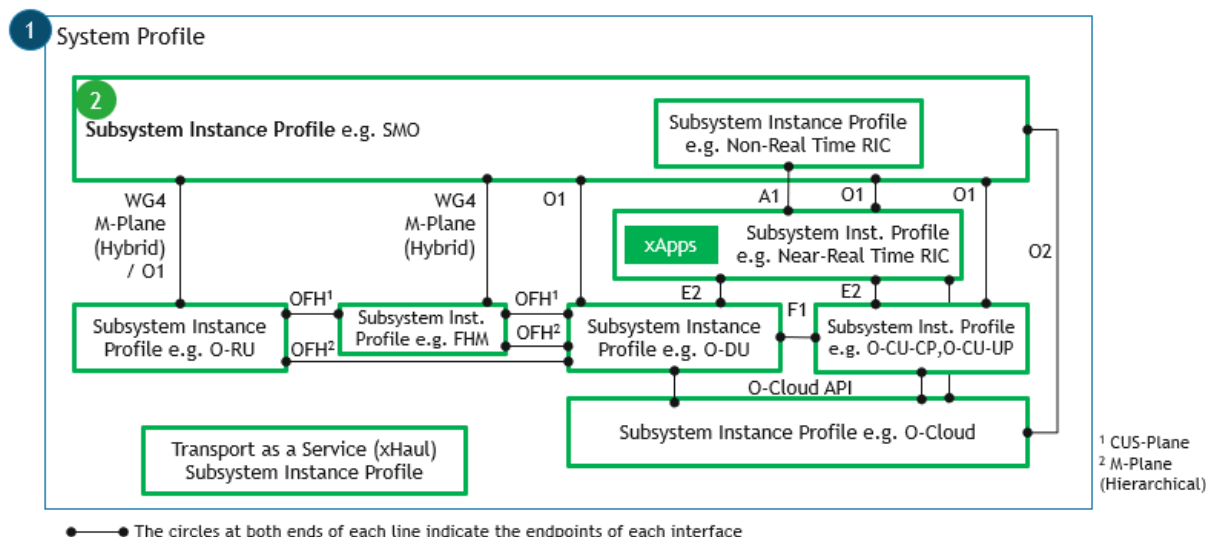


Figure 4-3: Scope of the Subsystem Instance Profile in relation to the Deployment Blueprint

Note that the intent of Figure 4-3 is used to show the scope of the Subsystem Instance Profile in relation to the O-RAN Deployment Blueprint. It is not intending to show the actual Subsystem Instance Profiles which must be included for all the O-RAN Deployment Blueprints.

OFH indicated in Figure 4-3 includes WG4 CUS and M plane implementation and for M-Plane both the Hierarchical and Hybrid models. Refer to [6][7] for more details.

A **Subsystem Instance** is defined to uniquely identify different implementations of a **Subsystem Type**.

For example, two O-RUs belonging to the **same Subsystem Type** (O-RU) which are optimally designed for Indoors versus Outdoors macro deployment scenarios will be considered as **different Subsystem Instances** as they would have different sets of capabilities and requirements defined for their respective purposes and therefore needing to be tested based on their respective sets of capabilities, requirements and KPIs.

The **Subsystem Instance Profile** is therefore dependent on the **Subsystem Type**.

The Subsystem Instance Profile should be specified to include the following information as defined in available technical and test specifications.

1. Features and capabilities
2. Functional specifications
3. Performance specifications (e.g. O-RU and O-DU sensitivity, QAM utilization, Spectral efficiency among others)
4. Performance measurements – Parameters with optional nominal values

4.3.1 Relationship between Subsystem Type and O-RAN network functions



Figure 4-4: Relationship between Subsystem Type and O-RAN Network Functions

The **Subsystem Type** is defined as the most fundamental building block / module of the O-RAN Deployment Blueprint which require robust interoperability testing as a Subsystem Instance in the O-RAN E2E testing framework. Therefore, the Subsystem Types may be different for different O-RAN Deployment Blueprints.

The relationship between Subsystem Type and O-RAN network functions can be described with the following scenarios

- Scenario 1: A single Subsystem Type composed with a single O-RAN network function ($M=N=1$)
- Scenario 2: A single Subsystem Type composed with more than a single O-RAN network function ($M=1, N>1$)
- Scenario 3: A single O-RAN network function maps to more than a single Subsystem Type ($M>1, N=1$)

Examples of a single Subsystem Type composed with more than a single O-RAN network function

1. Combined O-CU-CP and O-CU-UP implemented as an integrated Subsystem such as what is shown in cloud deployment scenarios A and B in [4]
2. Combined O-DU, O-CU-CP and O-CU-UP implemented as an integrated Subsystem such as what is shown in cloud deployment scenarios A and B in [4]
3. Combined O-RU and O-DU implemented as an integrated Subsystem such as what is shown in cloud deployment scenario E in [4]

Examples of a single O-RAN network function mapped to more than a single O-RAN Subsystem Type

1. O-Cloud can consist of multiple Subsystems including the hardware, Acceleration Abstraction Layer (AAL)
2. O-DU can consist of multiple Subsystems including L1 and L2 software stacks
3. Near-Real Time RIC can consist of one or more xApps running on the Near-Real Time RIC

Tables 4-1, 4-2 and 4-3 show the example listings of the Subsystem Types and their relationships to the corresponding O-RAN network functions for scenarios 1, 2 and 3, respectively.

Additional Subsystem Types which are not shown in these tables can be implemented with its own composition of the O-RAN network functions.

Table 4-1: Relationship of Subsystem Type to the O-RAN network functions ($M=N=1$)

Subsystem Type	O-RAN network function	Comments
O-RU	O-RU	
FHM	O-RU	FHM is applicable in Shared Cell (FHM mode) defined in WG4 specifications [6][7]. FHM is modelled as an O-RU with Open Fronthaul support (same as normal O-RU) and copy and combine function (additional to normal O-RU), but without radio transmission/reception capability [6][7].
O-DU	O-DU	
O-CU-CP	O-CU-CP	
O-CU-UP	O-CU-UP	
O-eNB	O-eNB	

Subsystem Type	O-RAN network function	Comments
O-Cloud	O-Cloud	
Near-Real Time RIC	Near-Real Time RIC	
Non-Real Time RIC	Non-Real Time RIC	
SMO	SMO	

Table 4-2: Relationship of Subsystem Type to the O-RAN network functions (M=1, N>1)

Subsystem Type	O-RAN network functions	Reference to [4] as appropriate
Integrated O-CU-CP, O-CU-UP [NR]	O-CU-CP, O-CU-UP	Integrated O-CU-CP, O-CU-UP can be implemented as a Subsystem composed of O-CU-CP, O-CU-UP as shown in Scenarios A, B etc.
Integrated O-RU, O-DU [NR / LTE]	O-RU, O-DU	O-RU, O-DU can be implemented as a single Subsystem as shown in Scenario E for NR.
Integrated O-DU, O-CU-CP, O-CU-UP [NR]	O-DU, O-CU-CP, O-CU-UP	O-DU, O-CU-CP, O-CU-UP can be implemented as a single Subsystem as shown in Scenarios A, B etc.
Integrated O-CU-CP, O-CU-UP, Near-Real Time RIC	O-CU-CP, O-CU-UP, Near-Real Time RIC	Near-Real Time RIC, O-CU-CP, O-CU-UP can be implemented as a single Subsystem as shown in Scenario C etc.
O-eNB [LTE]	O-RU, O-DU, CU	O-eNB can be implemented as a single O-RAN network function (O-eNB) or composed with LTE O-RU, O-DU, CU.

Table 4-3: Relationship of Subsystem Type to the O-RAN network functions (M>1, N=1)

Subsystem Type	O-RAN network function	Comments
Hardware Accelerator with Acceleration Abstraction Layer (AAL) APIs	O-Cloud	Hardware Accelerator with Open API allowing NFs applications to discover, configure, select and use (one or more) acceleration functions provided by a given accelerator on the cloud platform.
xApps	Near-Real Time RIC	The xApp is designed to run on the near-RT RIC which provides Open APIs for A1, E2, Management, Control and Shared Data Layer (SDL).
O-DU L1 stack	O-DU	High Physical layer (L1) stack in O-DU which provides API's to communicate with MAC Layer (L2) stack of O-DU.
O-DU L2 stack	O-DU	MAC Layer (L2) stack in O-DU which provides API's to communicate with High Physical Layer (L1) stack and RLC Layer stack of O-DU.

4.4 Subsystem Pair Interoperability Profiles

Each pair of Subsystem Instances and the Open Interface between them can be described in the O-RAN Deployment Blueprint with its Subsystem Pair Interoperability Profile.

Figure 4-5 shows the scope of the Subsystem Pair Interoperability Profile in relation to the O-RAN Deployment Blueprint.

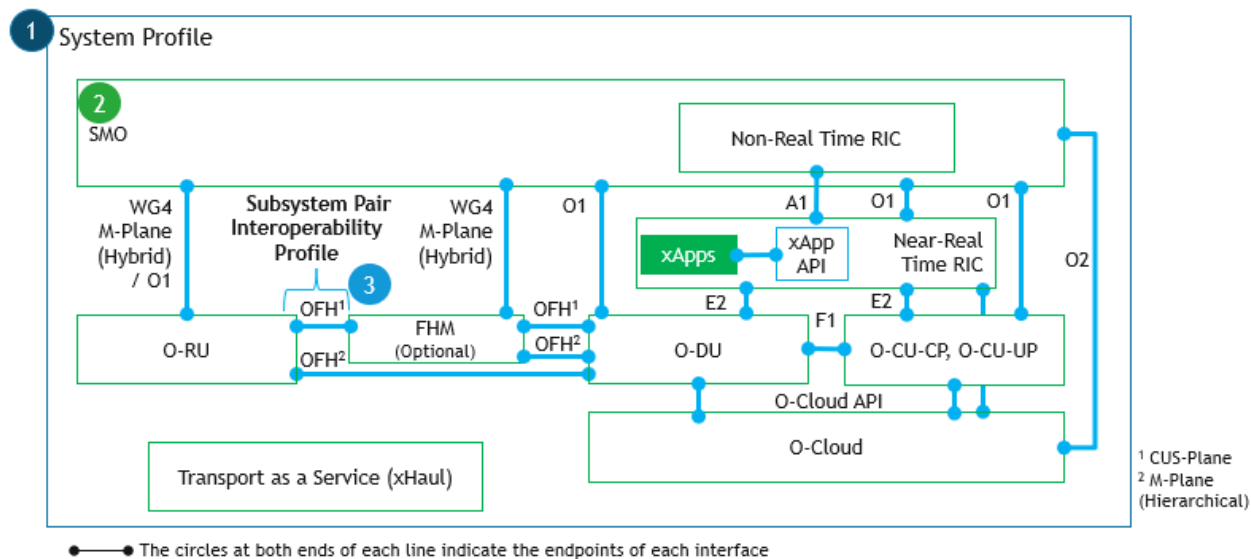


Figure 4-5: Scope of the Subsystem Pair Interoperability Profile in relation to the Deployment Blueprint

Note that the intent of Figure 4-5 is used to show the scope of the Subsystem Pair Interoperability Profile in relation to the O-RAN Deployment Blueprint. It is not intending to show the actual Subsystem Pair Interoperability Profile(s) which must be included for all the O-RAN Deployment Blueprints.

OFH indicated in Figure 4-5 includes WG4 CUS and M plane implementation and for M-Plane both the Hierarchical and Hybrid models. Refer to [6][7] for more details.

The Subsystem Pair Interoperability Profile is dependent on the pair of Subsystem Instances which are connected via an **Interface under Test (IUT)**.

Subsystem Pair Interoperability Profile defined in the O-RAN Deployment Blueprint includes the Interoperability Test (IOT) Profile for a specific Open Interface implementation between two Subsystem Instances.

The Subsystem Pair Interoperability Profile should be specified to include the following information as defined in available O-RAN WGs technical and test specifications

1. Features and capabilities
2. Functional specifications
3. Performance specifications
4. Performance measurements – Parameters with optional nominal values

Table 4-4 shows the mapping of Subsystem Pair Interoperability Profile to the respective O-RAN WGs.

Table 4-4: Subsystem Pair Interoperability Profile and respective O-RAN WGs

Subsystem Pair Interoperability Profiles	O-RAN WG(s)	O-RAN WG focus
Open Fronthaul (OFH)	WG4	WG4 IOT profiles C/U/S/M Planes
X2	WG5	WG5 C and U Plane Profiles
Xn	WG5	WG5 C and U Plane Profiles
E1	WG5	WG5 C Plane Profiles (currently not available) Note: E1 not supported by existing WG5 specifications
F1-C/F1-U	WG5	WG5 C and U Plane Profiles
E2	WG3	
A1	WG2	

Subsystem Pair Interoperability Profiles	O-RAN WG(s)	O-RAN WG focus
O1	WG1	
O2	WG6	
xApp (Near-RT RIC)	WG3	Open APIs for xApp
AAL	WG6	

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Chapter 5 O-RAN End-to-End (E2E) Testing Framework

This chapter describes the E2E testing methodology for the holistic testing and evaluation of the O-RAN Deployment Blueprints ensuring that these Blueprints can be robustly tested and optimized, ensuring consistent high quality and performance in a deterministic, repeatable and reproducible manner across the entire technology lifecycle.

Figure 5-1 shows the major phases of the technology lifecycle which will require testing and optimization.

In this version of the specification, the focus will be on the Lab Verification phase.

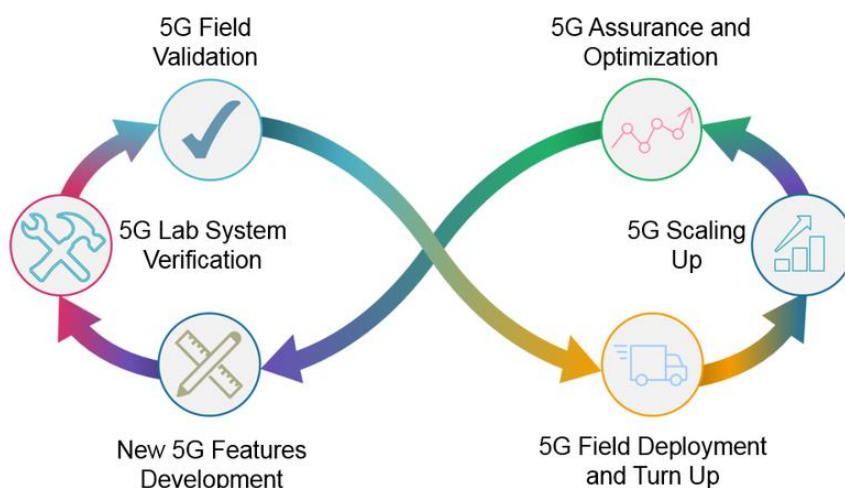


Figure 5-1: Major Stages of the Technology Lifecycle

The proposed testing methodology involves Subsystem testing, Subsystem Pairing and Open Interfaces Interoperability testing and E2E System testing.

5.1 Stages and Sequence of Testing

Figure 5-2 shows the stages and sequence of testing which can be performed on the O-RAN Deployment Blueprint

1. Perform Subsystem Testing on each of the Subsystems defined in the Blueprint
2. Perform Interoperability testing with pairs of Subsystems which have passed the Subsystem Testing stage
3. Perform E2E System Testing after all Subsystem Pairs passed the Subsystem Pair Interoperability Testing stage

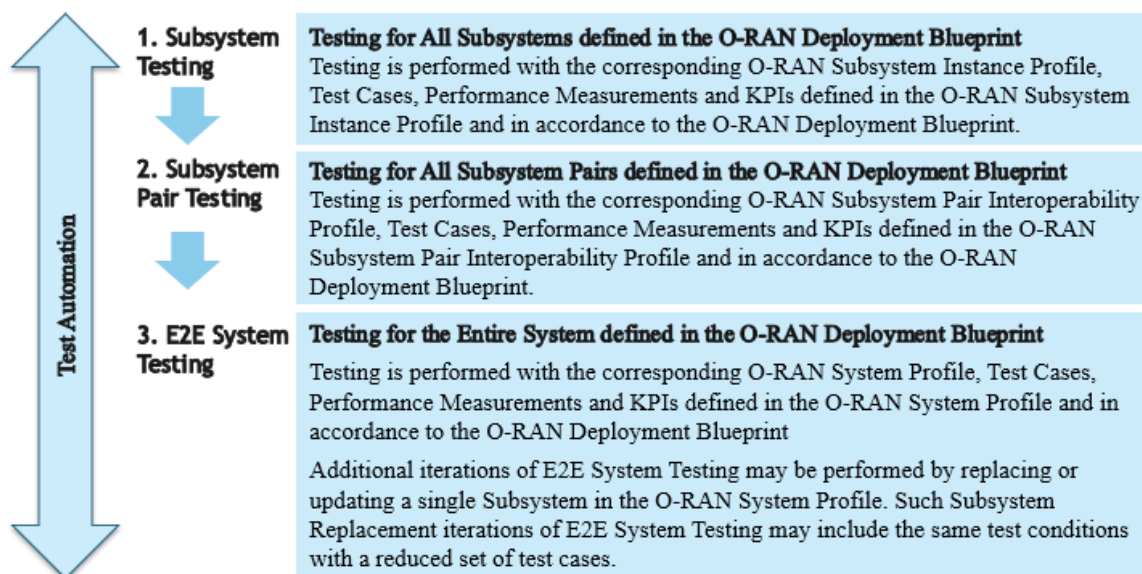


Figure 5-2: Stages and Sequence of Testing to be performed on the O-RAN Deployment Blueprint

These various stages of testing can be performed (or repeated) for any modifications to the O-RAN Deployment Blueprint which can include one or more of the following conditions

1. New Subsystem vendors
2. New releases of the software, firmware and/or hardware in any of the Subsystems
3. Configuration changes to the software, firmware and/or hardware in any of the Subsystems

The modifications may or may not be accompanied with corresponding updates in the System profile, Subsystem Instance profiles, Subsystem Pair and Open Interfaces Interoperability profiles and therefore the test cases, performance measurements and KPIs.

The modifications are required to be thoroughly reviewed to be able to evaluate the impacts of these modifications to the corresponding modified Subsystem Instance and the Subsystems Pair and E2E System associated with the modified Subsystem Instance. The impact evaluation can then be applied to the design and selection of test cases within each of the Test Stages needing to be executed due to the modifications.

For example, a new release can be labelled as a major or minor release but it is only through detailed analysis of the modifications in this new release can help to determine the specific impacts and therefore test cases within each of the Test Stages needing to be executed for this new release.

Subsystem Replacement Testing is an implementation variant of E2E System Testing with one of Subsystems replaced in the E2E System Test environment while keeping the other Subsystems and all the test configurations the same. Subsystem Replacement Testing methodology is further discussed in Section 5.4.1.

Test Automation will be required to make the testing procedure extremely efficient as the Test Automation framework can be designed and used to handle repetitive, time-consuming tests which are required to be executed when any part of the Blueprint has been modified. This results in improved accuracy as automated tests perform the same steps precisely every time they are executed and create detailed reports. Test Automation is expected to be applied to all Stages of Testing shown in Figure 5-2.

Figure 5-3 shows the scope and relationship of Subsystem Testing, Subsystem Pairing and Open Interfaces testing and E2E System Testing.



3 Few notes for clarification purposes on Figure 5-3

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5.2 Subsystem Testing

Subsystem Testing on each of the Subsystems defined in the O-RAN Deployment Blueprint is typically performed and passed before Interoperability Testing is to be performed for each of the Subsystem Pairs as defined in the Blueprint.

Subsystem testing should be performed with the requirements of the Subsystem specified in the respective Subsystem Instance Profile and Test Cases and KPIs defined by the WGs responsible for the specification of the Subsystem. Noting that each Subsystem can have Requirements, Test Cases and KPIs specified by one or more WGs.

Figure 5-5 shows a summarized view of the Subsystem, Subsystem Pair and Open Interfaces and the WGs responsible for the specification of each the Subsystem and Open Interfaces between the Subsystem Pairs.

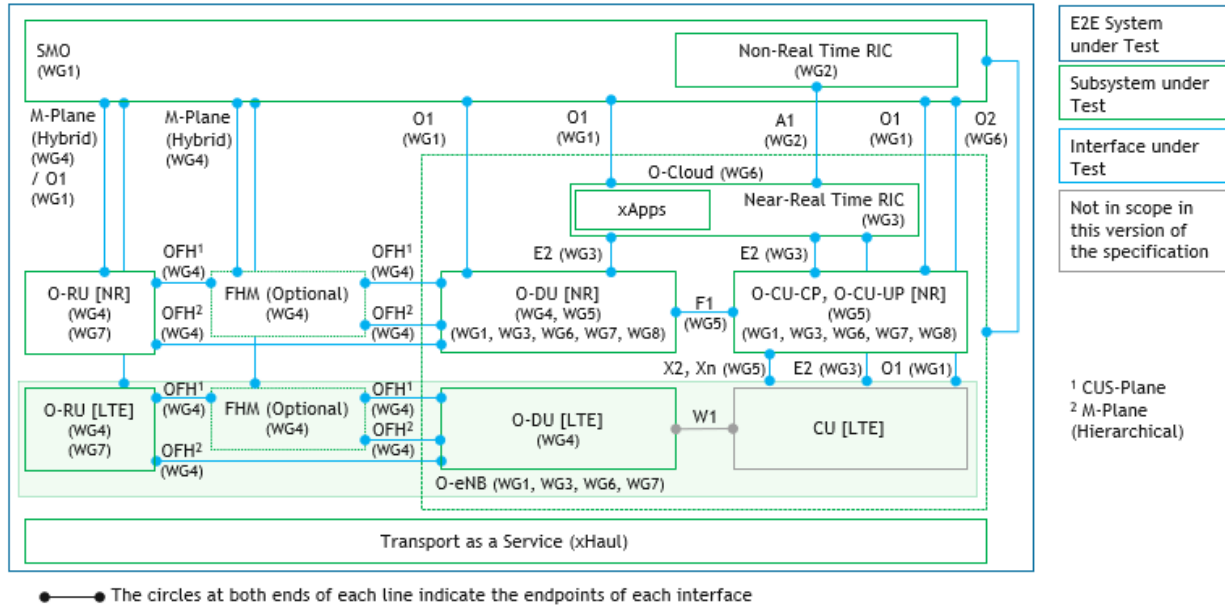


Figure 5-5: Summarized view of the Subsystem, Subsystem Pair Open Interfaces and the responsible WGs

OFH indicated in Figure 5-5 includes WG4 CUS and M plane implementation and for M-Plane both Hierarchical and Hybrid models. Refer to [6][7] for more details.

Table 5-1 shows the relationship between each of the Subsystem Types and the WGs responsible for various specifications aspects of these Subsystems.

Table 5-1: O-RAN Subsystems and WGs responsible for specifications

Subsystem Type	Subsystem Implementation Option	WG and Specifications
O-RU	Separated/Integrated O-DU and O-RU Management Plane – Hierarchical and Hybrid modes	WG4: OFH CUS and M-plane specifications in cases where the O-DU and O-RU are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Shared Cell (FHM or Cascade mode)	WG4: OFH specifications (CUS & M plane)
	Whitebox	WG7
	Management Plane – Hybrid mode (WG4 M Plane Hybrid mode, O1 alignment)	WG4: M-Plane specification; WG1: O1
	Location (Cell Site)	WG6: Cell Site (if VNF/CNF) (for further study FFS)
	Software and Hardware decoupling	WG6 (for further study FFS)
FHM	Shared Cell (FHM mode)	WG4: OFH specifications (CUSM plane)
O-DU	Management Plane	WG1: O1; WG5: O1; WG4: M-Plane specification (if the O-DU is not integrated with the O-RU)

Subsystem Type	Subsystem Implementation Option	WG and Specifications
	Separated/Integrated O-DU and O-RU	WG4: OFH CUS and M-plane specifications in cases where the O-DU and O-RU are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Separated/Integrated O-DU and O-CU-CP/O-CU-UP	WG5: F1 C and U plane profiles where the O-DU and O-CU-CP/O-CU-UP are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Software and Hardware decoupling	WG6
	Reference Stack (PHY Hi, MAC/RLC)	WG8: O-DU Software Architecture and APIs specification
	Accelerator decoupling (AAL)	WG6
	PNF, VNF, CNF	WG6: VNF, CNF
	Location (Cell, Edge site)	WG6: VNF/CNF
	Whitebox	WG7
	Near-Real time RIC support (counters and stats perspective, E2)	WG3: E2; O-DU support for counters and stats perspective should be part of WG5
Integrated O-CU-CP, O-CU-UP ¹	Base functions	WG1: Management Plane (O1) WG5: O1 information and data models
	Separated/Integrated O-DU and O-CU-CP/O-CU-UP	WG5: F1 C and U plane profiles where the O-DU and O-CU-CP/O-CU-UP are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Software and Hardware decoupling	WG6
	Reference Stack (PDCP, RRC, SDAP – in SA mode)	WG8: O-CU-CP/O-CU-UP Software Architecture and APIs specification
	Accelerator decoupling (AAL)	WG6
	PNF, VNF, CNF	WG6: VNF, CNF
	NR Deployment Options (NSA Option 3x, SA Option 2, others)	WG5: X2 (NSA), Xn (SA)
	Location (Edge, Regional site)	WG6: VNF/CNF
	Whitebox	WG7
O-CU-CP	Base functions	WG5: F1-C; E1 WG1: Management Plane (O1) WG5: O1 information and data models Note: E1 not supported by existing WG5 specifications
	Software and Hardware decoupling	WG6
	Reference Stack (PDCP, RRC)	WG8: O-CU-CP/O-CU-UP Software Architecture and APIs specification
	Accelerator decoupling (AAL)	WG6
	PNF, VNF, CNF	WG6: VNF, CNF
	NR Deployment Options (NSA Option 3x, SA Option 2, others)	WG5: X2 (NSA), Xn (SA)
	Location (Edge, Regional site)	WG6: VNF/CNF
	Whitebox	WG7

Subsystem Type	Subsystem Implementation Option	WG and Specifications
	Near-Real time RIC support (counters and stats perspective, E2)	WG3: E2; O-CU-CP/O-CU-UP support for counters and stats perspective should be part of WG5
O-CU-UP	Base functions	WG5: F1-U; E1 WG1: Management Plane (O1) WG5: O1 information and data models Note: E1 not supported by existing WG5 specifications
	Software and Hardware decoupling	WG6
	Reference Stack (PDCP, SDAP – in SA mode)	WG8: O-CU-CP/O-CU-UP Software Architecture and APIs specification
	Accelerator decoupling (AAL)	WG6
	Management Plane (O1)	WG1: O1
	PNF, VNF, CNF	WG6: VNF, CNF
	NR Deployment Options (NSA Option 3x, SA Option 2, others)	WG5: X2 (NSA), Xn (SA)
	Location (Edge, Regional site)	WG6: VNF/CNF
	Whitebox	WG7
	Near-Real time RIC support (counters and stats perspective, E2)	WG3: E2; O-CU-CP/O-CU-UP support for counters and stats perspective should be part of WG5
O-eNB	Base functions	WG1: Management Plane (O1)
	Separated/Integrated O-DU and O-RU Management Plane – Hierarchical and Hybrid modes	WG4: OFH CUS and M-plane specifications in cases where the O-DU and O-RU are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Software and Hardware decoupling	WG6
	Accelerator decoupling (AAL)	WG6
	PNF, VNF, CNF	WG6: VNF, CNF
	NR Deployment Options (NSA Option 3x)	WG5: X2 (NSA)
	Location (Cell, Edge, Regional site)	WG6: VNF, CNF
	Whitebox	WG7
O-Cloud	Base functions	WG6: O2
	Software and Hardware decoupling	WG6
	Accelerator decoupling (AAL)	WG6
	Location (Cell, Edge, Regional site)	WG6: VNF, CNF
Near-Real Time RIC	Base functions	WG3: E2, xApp; WG2: A1
	PNF, VNF, CNF	WG6: VNF, CNF
	Location (Edge, Regional site)	WG6: VNF, CNF
	xApps	WG3
Non-Real Time RIC	Base functions	WG2: A1
	PNF, VNF, CNF	WG6: VNF, CNF
	Location (Edge, Regional site)	WG6: VNF, CNF
SMO	Base functions	WG1: O1, O2

Subsystem Type	Subsystem Implementation Option	WG and Specifications
xApp	Base functions	WG3
¹ Refer to Table 4-2 for the mapping of Subsystem Type to O-RAN network functions		

Performing Subsystem Testing on the O-DU for example will require Technical and Test specifications from

1. WG1 and WG5 for O1 Management Plane support (WG1 specifies the management services provided over the O1 interface, WG5 specifies the information/data models referencing 3GPP specifications where required)
2. WG4 Open Fronthaul (OFH) CUS and M Planes (if the O-DU is not integrated with the O-RU)
3. WG5 Open F1 (if the O-DU is not integrated with the O-CU-CP and O-CU-UP)
4. WG6 Cloud Platform (if the O-DU is implemented as VNF/CNF running on a Cloud Platform)
5. WG8 Software Architecture and APIs (if the O-DU is implemented using multi-vendors Reference stacks)
6. WG6 Accelerator Abstraction Layer (AAL) (if the O-DU is implemented with Accelerator decoupling)
7. WG3 Near-Real time RIC support (if E2 is to be implemented)

Performing Subsystem Testing on the O-RU for example will require Technical and Test specifications from

1. WG4 Open Fronthaul (OFH) CUS and M Planes (if the O-RU is not integrated with the O-DU)
2. WG4 Open Fronthaul (OFH) CUS and M Planes (if the O-RU is configured in Shared Cell operation in either FHM or Cascade mode)
3. WG1 and WG4 for O1 Management Plane support (WG1 specifies the management services provided over the O1 interface, WG4 specifies the information/data models referencing 3GPP specifications where required. Noting that O1 support for O-RU is still work in progress)
4. WG7 Whitebox reference design (if the O-RU is implemented based on whitebox reference design)
5. WG4 Near-Real time RIC support (if performance counters and statistics are required to support use cases enabled through Near-Real time RIC)

One of the key objectives of Subsystem testing is to validate the functionality of production grade DUT when the DUT is not operating in test mode and in few cases in standardized test mode as when required. One example of standardized test mode can be 3GPP test modes.

Hence it is important to ensure that the DUT is not negatively impacted with the utilization of internal functions solely to support Subsystem testing, i.e., DUT is not expected to be testing tools when deployed in production networks and therefore DUT should not be used as testing tool during Subsystem tests.

Subsystem tests are performed with a set of testing tools which are used to both apply active stimulus and as well as passive monitoring and measurements of the DUTs.

It is recommended for Subsystem testing to adopt a wraparound testing approach for each of the Subsystem as the Device under Test (DUT) and Interfaces under Test (IUT), therefore providing a testing framework for comprehensive evaluation and testing prior to Multi-vendors Interoperability (MV-IOT) Subsystem pairing and Open interfaces testing.

The wraparound testing approach involves emulating or using reference test platforms for the relevant Subsystems surrounding the Subsystem (DUT) as per defined in the O-RAN Deployment Blueprint. Reference test platforms can be Subsystems designed for commercial purpose or test applications which are used to support the wraparound test setup. For example, an O-Cloud reference test platform can be used to support wraparound testing of the virtualized/cloudified O-RAN network function designed to operate on the O-Cloud Subsystem.

This would obviously include the required Service Management and Orchestration (SMO) and necessary procedures to bring the DUT into Operational state.

Table 5-2 shows the relationship between each of the Subsystem Types and the surrounding Subsystems which are recommended to be emulated or using reference test platforms as part of the wrap around testing approach.

Table 5-2: Subsystem Types, Implementation Options and Surrounding Subsystems

Subsystem Type	Subsystem Implementation Option	Surrounding Subsystems which are recommended to support wraparound testing
O-RU	Separated/Integrated O-DU and O-RU	O-DU in cases where the O-DU and O-RU are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Shared Cell (FHM)	FHM for CUS planes and NETCONF Client in O-DU/NMS for M plane
	Shared Cell (Cascade mode)	For O-RU #1..#N in the cascaded chain. O-DU in the case if O-RU #1 (DUT) is first in the chain closest to the O-DU; O-RU #N-1 in the case if O-RU #N (DUT) is last in the chain; For $N > 2$ and $X < N$, O-RU #X-1 and O-RU #X+1 in the case if O-RU #X (DUT)
	Whitebox	Reference Hardware Platform
	Cloud Platform	Reference Cloud Platform (For further study FFS)
FHM	Shared Cell (FHM mode)	One or more O-RUs and One O-DU
O-DU	Management Plane	SMO (O1)
	Separated/Integrated O-DU and O-RU	One or more O-RUs in cases where the O-DU and O-RU are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Separated/Integrated O-DU and O-CU-CP/O-CU-UP	O-CU-CP/O-CU-UP (F1-C/F1-U) in cases where the O-DU and O-CU-CP/O-CU-UP are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Separated O-CU-CP and O-CU-UP	O-CU-CP (F1-C) and O-CU-UP (F1-U)
	Shared Cell (FHM)	FHM for CUS and M Plane. One or more O-RUs (NETCONF Server) for M-Plane
	Shared Cell (Cascade mode)	O-RU #1
	Whitebox	Reference Hardware Platform
	Cloud Platform	Reference Cloud Platform
	Accelerator decoupling (AAL)	AAL
	Near-Real time RIC support	Near-Real time RIC (E2)
Integrated O-CU-CP, O-CU-UP ¹	Management Plane	SMO (O1)
	Separated/Integrated O-DU and O-CU-CP/O-CU-UP	One or more O-DUs (F1) in cases where the O-DU and O-CU-CP/O-CU-UP are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Whitebox	Reference Hardware Platform
	Cloud Platform	Reference Cloud Platform
	Accelerator decoupling (AAL)	AAL
	NR Deployment Option (NSA Option 3x, 3a)	eNB (X2)
	NR Deployment Option (SA Option 2)	gNB (Xn)
	Near-Real time RIC support	Near-Real time RIC (E2)
O-CU-CP	Management Plane	SMO (O1)
	Separated O-CU-CP and O-CU-UP	One or more O-CU-UP (E1) Note: E1 not supported by existing WG5 specifications
	Separated/Integrated O-DU and O-CU-CP	One or more O-DUs (F1-C) in cases where the O-DU and O-CU-CP are implemented as separate nodes and

Subsystem Type	Subsystem Implementation Option	Surrounding Subsystems which are recommended to support wraparound testing
		they can either be co-located in the same or distributed locations
	Whitebox	Reference Hardware Platform
	Cloud Platform	Reference Cloud Platform
	Accelerator decoupling (AAL)	AAL
	NR Deployment Option (NSA Option 3x, 3a)	eNB (X2)
	NR Deployment Option (SA Option 2)	gNB (Xn)
	Near-Real time RIC support	Near-Real time RIC (E2)
O-CU-UP	Management Plane	SMO (O1)
	Separated O-CU-CP and O-CU-UP	O-CU-CP (E1) Note: E1 not supported by existing WG5 specifications
	Separated/Integrated O-DU and O-CU-UP	One or more O-DUs (F1-U) in cases where the O-DU and O-CU-UP are implemented as separate nodes and they can either be co-located in the same or distributed locations
	Whitebox	Reference Hardware Platform
	Cloud Platform	Reference Cloud Platform
	Accelerator decoupling (AAL)	AAL
	NR Deployment Option (NSA Option 3x)	eNB (X2)
	NR Deployment Option (SA Option 2)	gNB (Xn)
	Near-Real time RIC support	Near-Real time RIC (E2)
O-eNB	Management Plane	SMO (O1)
	Whitebox	Reference Hardware Platform
	Cloud Platform	Reference Cloud Platform
	NR Deployment Option (NSA Option 3x)	eNB (X2)
	Near-Real time RIC support	Near-Real time RIC (E2)
O-Cloud	Management Plane	SMO (O2)
	Cloud Platform	O-RAN network functions operating on Cloud platforms – O-DU, O-CU-CP, O-CU-UP, O-eNB, Near-Real time RIC, Non-Real time RIC, SMO
	Accelerator decoupling (AAL)	AAL
Near-Real Time RIC	Management Plane	SMO (O1)
	Cloud Platform	Reference Cloud Platform
	E2 Nodes	E2 Nodes (E2) – O-DU, O-CU-CP, O-CU-UP, O-eNB
	xApps	xApps
	Non-Real time RIC support	Non-Real time RIC (A1)
Non-Real Time RIC	Management Plane	SMO (O1)
	Cloud Platform	Reference Cloud Platform
	Near-Real time RIC support	Near-Real time RIC (A1)
SMO	Management Plane	O-RAN Managed Element (O1)
xApp	Near-Real time RIC support	Near-Real time RIC Platform and E2 Nodes (E2) - O-DU, O-CU-CP, O-CU-UP, O-eNB, Non-Real time RIC (A1)

¹ Refer to Table 4-2 for the mapping of Subsystem Type to O-RAN network functions

1. One or more O-RUs (Open Fronthaul OFH) in cases where the O-DU and O-RUs are implemented as separate nodes
2. O-CU-CP/O-CU-UP (F1-C/F1-U) in cases where the O-DU and O-CU-CP/O-CU-UP are implemented as separate nodes
3. In Separated O-CU-CP and O-CU-UP configuration - O-CU-CP (F1-C) and O-CU-UP (F1-U)
4. In Shared Cell (FHM) configuration - FHM for CUS and M Plane (Open Fronthaul OFH). One or more O-RUs (NETCONF Server) for M-Plane
5. In Shared Cell (Cascade) configuration - O-RU #1 (Open Fronthaul OFH)
6. Reference Cloud Platform
7. Near-Real time RIC (E2)
8. SMO (O1)

Figure 5-6 shows an example Wrap around testing setup with the O-DU configured in NR SA mode as the Subsystem under Test.



OFG indicated in Figure 5-6 includes WG4 CUS and M plane implementation and for M-Plane both Hierarchical and Hybrid models. Refer to [6][7] for more details.

Subsystem Pairing and Open Interfaces Interoperability testing is typically performed after the Subsystem testing on each of the Subsystems involved in the Pairing has been successfully completed and prior to E2E System Testing on all the Subsystems and Subsystems Pairs defined in the O-RAN Deployment Blueprint.

Subsystem Pairing and Open Interfaces testing should be performed with the requirements of the Subsystem Pair and Open Interfaces specified in the respective Subsystem Pair Interoperability Profiles and Test Cases and KPIs defined by the WGs responsible for the specification of the Open Interfaces and Interoperability Test (IOT) profiles.

Refer to the WGs' Technical and Test specifications for the respective Open Interfaces shown in Table 3-2 for more details on the recommended Subsystem Pairing and Open Interfaces testing approach and detailed procedures.

One of the key objectives of Subsystem Pairing and Open Interfaces testing is to validate the functionality of production grade DUTs when all the DUTs are not operating in test modes. Hence it is important to ensure that the DUTs are not negatively impacted with the utilization of internal functions solely to support interoperability testing, i.e., DUTs are not expected to be testing tools when deployed in production networks and therefore DUTs should not be used as testing tools during interoperability tests.

Interoperability tests are performed with a set of testing tools which are used to both apply active stimulus and as well as passive monitoring and measurements of the DUTs.

It is recommended for Subsystem Pairing and Open Interfaces Interoperability testing to adopt a wraparound testing approach for each of the Subsystem Pair as the DUTs and Open Interface as the IUT, therefore providing a testing framework for comprehensive evaluation & testing prior to E2E System testing.

The wraparound testing approach involves emulating or using reference test platforms for the relevant Subsystems surrounding the Subsystem Pair as the DUTs and Open Interface as the IUT as per defined in the O-RAN Deployment Blueprint. Reference test platforms can be Subsystems designed for commercial purpose or test applications which are used to support the wraparound test setup. For example, an O-Cloud reference test platform can be used to support wraparound testing of the virtualized/cloudified O-RAN network function designed to operate on the O-Cloud Subsystem.

This would obviously include the required Service Management and Orchestration (SMO) and necessary procedures to bring the DUTs into Operational state.

Table 5-3 shows the relationship between each of the Subsystem Pair and Open Interfaces and the surrounding Subsystems which are recommended to be emulated or using reference test platforms as part of the wrap around testing approach.

Table 5-3: O-RAN Subsystem Pair and Open Interfaces and the surrounding Subsystems

O-RAN Open Interfaces	O-RAN Subsystem Pairs	Surrounding Subsystems which are recommended to support wraparound testing – if implemented (Interface)
A1	Non-RT RIC function and Near-RT RIC function	E2 nodes (E2); SMO (O1) O-Cloud reference platform (O-Cloud API)
E1	O-CU-CP and O-CU-UP	O-DU (F1); near-RT-RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) Note: E1 not supported by existing WG5 specifications
E2	Near-RT RIC function and E2 Nodes for E2 functions	Non-RT-RIC function (A1); SMO (O1) O-Cloud reference platform (O-Cloud API) If E2 node is O-CU-CP/O-CU-UP: O-DU (F1-C/F1-U, E2) If E2 node is O-DU: O-CU-CP/O-CU-UP (F1-C/F1-U, E2); O-RU (OFH); FHM (OFH) If E2 node is O-eNB: no additional requirements
F1-C	O-CU-CP and O-DU	O-CU-UP (E1); O-RU (OFH); FHM (OFH); near-RT-RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) Note: E1 not supported by existing WG5 specifications
F1-U	O-CU-UP and O-DU	O-CU-CP (E1); O-RU (OFH); FHM (OFH); near-RT-RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) Note: E1 not supported by existing WG5 specifications

O-RAN Open Interfaces	O-RAN Subsystem Pairs	Surrounding Subsystems which are recommended to support wraparound testing – if implemented (Interface)
O1	O-RAN Managed Element and Management entity (SMO)	
O2	SMO and O-Cloud	Management plane interface between SMO and O-Cloud
Open Fronthaul (OFH) CUS and M-Plane (Hierarchical and Hybrid Modes)	O-DU and O-RU	O-CU-CP (F1-CP); O-CU-UP (F1-UP); near-RT-RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) at the O-DU
Open Fronthaul (OFH) M-Plane (Hybrid Mode)	SMO and O-RU	O-DU (OFH)
X2-C	eNB and en-gNB	Near-Real Time RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) at the O-CU-CP/O-CU-UP and O-DU
X2-U	eNB and en-gNB	Near-Real Time RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) at the O-CU-CP/O-CU-UP and O-DU
Xn-C	eNB and ng-gNB	Near-Real Time RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) at the O-CU-CP/O-CU-UP and O-DU
Xn-U	eNB and ng-gNB	Near-Real Time RIC function (E2); SMO (O1) O-Cloud reference platform (O-Cloud API) at the O-CU-CP/O-CU-UP and O-DU

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2 Performing Subsystem Pairing and Open Interfaces Interoperability testing using the wraparound test approach on the O-
3 RU and O-DU for example may require emulating or using reference test platforms for the following surrounding O-RAN
4 network functions

- 5 1. O-CU-CP (F1-C) and O-CU-UP (F1-U) in cases where the O-DU and O-CU-CP/O-CU-UP are implemented as
- 6 separate nodes
- 7 2. In Separated O-CU-CP and O-CU-UP configuration - O-CU-CP (F1-C) and O-CU-UP (F1-U)
- 8 3. Reference Cloud Platform in cases where the O-RU and/or O-DU is implemented on Cloud Platforms
- 9 4. Near-Real time RIC (E2)
- 10 5. SMO (O1)

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12 Noting that the OFH and the corresponding Management interfaces are the IUTs in this test setup. The rest of the interfaces
13 which are used to connect the DUTs to the emulated network functions and/or reference test platforms are not the IUTs
14 but required to support the wrap around testing approach.

15 Refer to the Subsystem Technical and Test specifications shown in Table 3-2 for more details on the recommended
16 Subsystem Pairing and Open Interfaces Interoperability Testing approach and detailed procedures.

17 Figure 5-7 shows an example with the O-RU and O-DU configured in NR SA mode, as the Subsystem under Test and
18 surrounding Subsystems which are recommended to be emulated or using reference test platforms for wrap around
19 Subsystem testing approach.

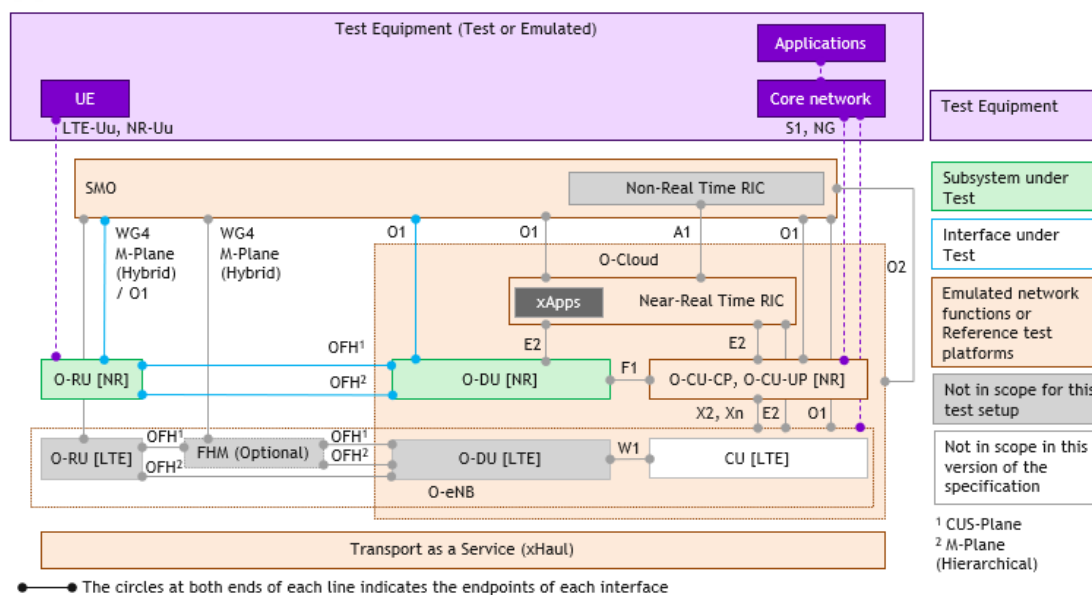


Figure 5-7: Wrap around testing setup for O-RU and O-DU as the Subsystem Pair under Test

O/FH indicated in Figure 5-7 includes WG4 CUS and M plane implementation and for M-Plane both Hierarchical and Hybrid models. Refer to [6][7] for more details.

5.4 E2E System Multi-vendors Interoperability Testing

Multi-vendors interoperability (MV-IOT) system testing enables validation and verification of the entire O-RAN Deployment Blueprint as a complete E2E system under test (SUT).

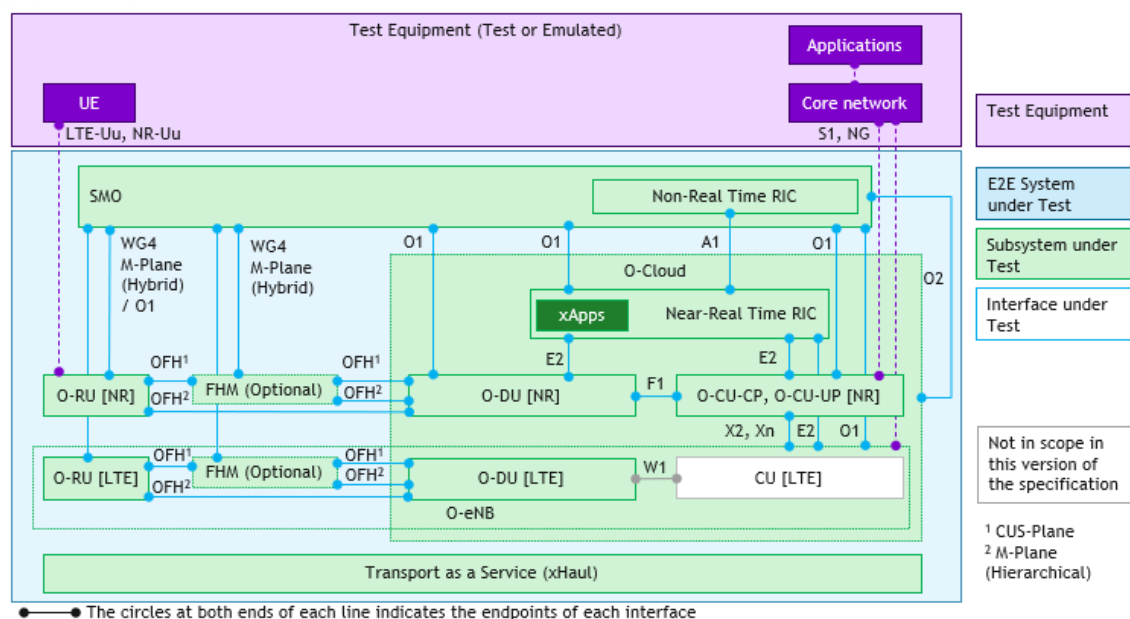
E2E System testing should be performed after Subsystem Testing, Subsystem Pair and Open Interfaces Interoperability Testing have been successfully completed for all the Subsystems and Subsystem Pairs defined in an O-RAN Deployment Blueprint.

One of the key objectives of E2E System testing is to validate the functionality of production grade DUTs when all the DUTs in the SUT are not operating in test modes. Hence it is important to ensure that the DUTs are not negatively impacted with the utilization of internal functions solely to support interoperability testing, i.e., DUTs are not expected to be testing tools when deployed in production networks and therefore DUTs should not be used as testing tools during E2E System tests.

E2E System tests are performed with a set of testing tools which are used to both apply active stimulus and as well as passive monitoring and measurements of the DUTs.

It is recommended for System testing to adopt a wraparound testing approach for all the Blueprint defined Subsystems as DUTs and Open interface connecting these Subsystems as IUTs, therefore providing a testing framework for comprehensive evaluation and testing.

The wraparound testing approach involves using Test Equipment such as the UEs (test and/or emulated), Applications/Services (test and/or emulated) and Core network (test and/or emulated) as shown in Figure 5-8 below.



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Figure 5-9 shows the wrap around testing setup for E2E System under Test with one of the Subsystems replaced. In this example, the O-DU subsystem is replaced as part of the E2E System Testing setup. Multiple vendors' O-DUs can be tested in the same E2E System Testing environment with the other Subsystems in the E2E test environment remaining the same.

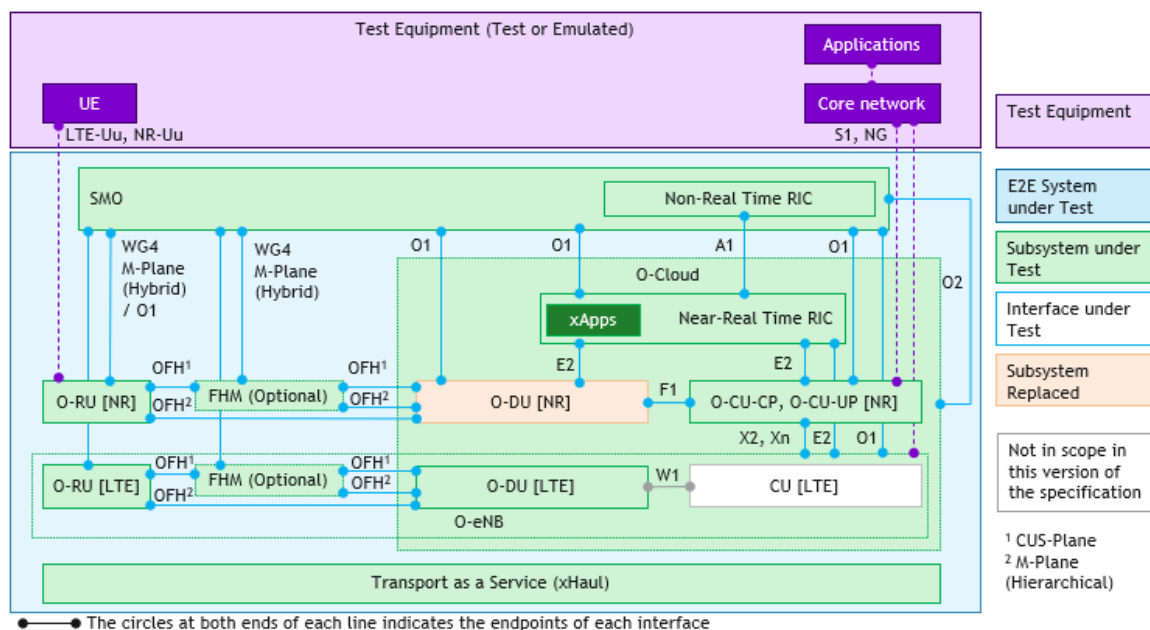


Figure 5-9: Wrap around testing setup for E2E System under Test (SUT) with one Subsystem Replaced

OFH indicated in Figure 5-9 includes WG4 CUS and M plane implementation and for M-Plane both Hierarchical and Hybrid models. Refer to [6][7] for more details.

Chapter 6 O-RAN E2E Test Cases and KPIs

6.1 Introduction

The O-RAN E2E testing framework aims to evaluate the entire O-RAN Deployment Blueprint holistically as a System under Test (SUT) based on a set of well-defined E2E Test Cases and Key Performance Indicators (KPIs).

The purpose of this Chapter is to outline the key considerations along with information to help guide testing and validating the O-RAN Deployment Blueprints with the relevant E2E Test Cases and KPIs.

The following considerations will be discussed in further details in the next sections in this Chapter

1. Use Cases and Services Requirements
2. O-RAN Deployment Blueprint dependencies
3. Relevance of 3GPP KPIs
4. Relevance of NGMN 5G Test Cases and KPIs defined in [10]
5. Test Conditions to be considered
6. Test Scenarios and KPIs to be considered

6.2 Use Cases and Services Requirements

The selection of the E2E Test Cases and KPIs are determined mainly by 2 factors

1. Use Cases and Services
2. O-RAN Deployment Blueprint which are optimized according to the Use Cases and Services requirements

Figure 6-1 shows a few examples for each of the major categories of use cases and services which are supported by 3GPP and O-RAN compliant networks.

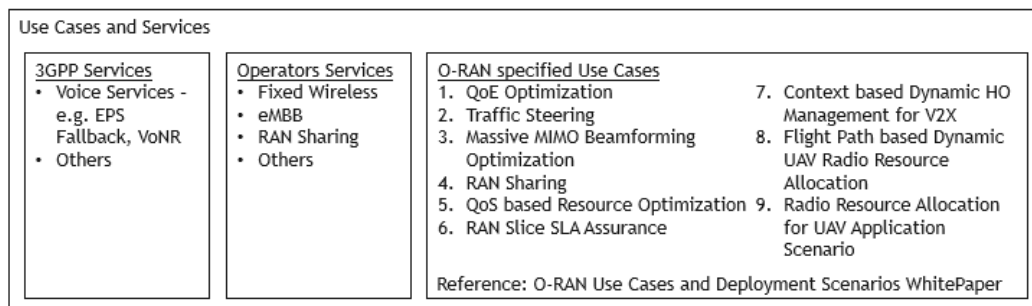


Figure 6-1: Examples of Use Cases/Services supported by 3GPP compliant and O-RAN RIC-enabled networks

To convince an operator to invest in O-RAN infrastructure, 3GPP and Operators' Services which can be supported with 3GPP compliant networks must be well supported with O-RAN compliant networks.

It can therefore be expected that operators will require comprehensive system testing and evaluation to be performed on the O-RAN Deployment Blueprints using the E2E Test Cases and KPIs which are defined for the 3GPP and Operators' Services that are of interest to them.

O-RAN Alliance has specified a set of prioritized use cases which can be only supported by O-RAN RIC-enabled networks. These use cases leverage the O-RAN architecture and demonstrates its unique benefits. Key benefits of O-RAN architecture include the ability to utilize machine learning systems and artificial intelligence back end modules to empower network intelligence through open and standardized interfaces / feeds in a multi-vendor network.

Operators will require guidance on the E2E Test Cases and KPIs which can be used to test and evaluate the performance for each of these RIC-enabled use cases in order to be able to quantify the benefits of implementing these use cases using O-RAN RIC-enabled networks as compared to networks without O-RAN RIC capabilities.

6.3 O-RAN Deployment Blueprint dependencies

An O-RAN Deployment Blueprint can consist of multiple Subsystems from possibly multiple vendors. The KPIs can be measured as E2E KPIs across any combinations of vendors and as well as Subsystems and Subsystems Pairs KPIs for the different Subsystems and the Open interfaces between the Subsystems respectively.

The specific Performance Measurements, KPIs and their respective values defined for Subsystem Testing, Subsystem Pair and Open Interfaces Testing and E2E System Testing will be dependent on the O-RAN Deployment Blueprints.

For example, from the E2E System Testing perspective, the beam forming system performance gain KPIs specified for outdoors macro cells would not be applicable for indoors small cells which may not require beam forming capabilities. Similarly, the beam forming performance gain measurements specified for the O-RUs which are designed and optimized for outdoors macro cells systems would not be applicable for O-RUs which are designed and optimized for indoors small cells.

It is therefore important to ensure that testing is performed for each of the

1. Subsystem Instances with their defined Subsystem Instance profiles and associated KPIs
2. Subsystem Instances pairs with their defined Subsystem Pairs Interoperability profiles and associated KPIs
3. System Deployment Blueprints with the defined E2E System KPIs

6.4 Relevance of the 3GPP KPIs

As highlighted in Section 6.2, operators will require validation of the O-RAN Deployment Blueprints that the overall 3GPP System characteristics is as expected.

In general, 3GPP defines two categories of RAN KPIs

1. Performance data for technology tests, trials, and evaluation such as in 3GPP TR 38.913.
2. Performance data for network monitoring, assessment, analysis, optimization, and assurance which will be further discussed in Section 6.4.1.

The same set of E2E KPI categories and KPIs which are defined for 3GPP compliant networks and services will be directly applicable for O-RAN compliant networks.

Additional considerations for the application of 3GPP defined KPIs for O-RAN Deployment Blueprint testing and verification will be discussed in Section 6.4.2.

6.4.1 3GPP KPIs defined for network monitoring, assessment, analysis, optimization and assurance

From the 3GPP system perspective, the E2E KPIs defined for network monitoring, optimization and assurance are typically defined for the following categories

1. Availability (e.g. cells ready to service connections)
2. Accessibility (e.g. connection establishment rates, attempts/failures, latency)
3. Integrity (e.g. throughput, latency, jitter, retransmission rates)
4. Mobility (e.g. intra/inter RATs handover rates, interruption times for control / user planes)
5. Reliability (e.g. amount of sent packets which are successfully delivered to the destination within the time constraint required by the target service, divided by the total number of sent packets)
6. Retainability (e.g. drop rates for PDN/PDU sessions, call session)
7. Utilization (e.g. resources utilization based on certain traffic conditions and profiles such as fronthaul, compute)
8. Energy Efficiency (e.g. power consumption based on certain traffic conditions and profiles)

3GPP TS 28.554 [9] has defined a set of E2E KPI for the 5G System (5GS) and network slicing including the NR RAN and 5G Core (5GC).

6.4.2 Applicability to RAN, RAN Slices, Users and Services

3GPP KPIs are typically defined for 3GPP networks (RAN and Core), E2E network slices and 3GPP hosted services such as 3GPP voice services with VoLTE, EPS Fallback, VoNR and others.

Additional important KPIs dimensions which are required to be considered for O-RAN Deployment Blueprint testing and verification include

1. RAN Slices even though 3GPP has not yet standardized RAN slicing, this is one of the O-RAN use cases
2. Users and Services QoS to be evaluated for Accessibility, Integrity, Mobility, Retainability
3. Users and Services QoE to be evaluated

6.5 Relevance of the NGMN 5G Test Cases and KPIs

NGMN has developed a set of E2E Test Cases and KPIs which are designed to evaluate the performance of 3GPP 5G NR networks [10] with its work on the definition of the testing framework for NGMN's Pre-Commercial 5G networks trials initiative.

The current published version of this NGMN document focuses on outdoors field testing of the 5G NR system configured in NSA EN-DC mode, which includes validation of the 5G NR radio interface, E2E KPIs and features which require E2E validation such as the E2E latency and E2E network slicing.

This set of NGMN defined E2E Test Cases and KPIs can be leveraged to a large extent for testing and validating the O-RAN Deployment Blueprints in both the lab and field-testing environments.

Table 6-1 lists the E2E Test Cases and KPIs documented in [10] which should be referred to for more details.

Table 6-1: NGMN Pre-Commercial 5G networks trials E2E Test Cases and KPIs [10]

Test Case focus	KPIs and Test Scenarios
Trial test requirements	
Latency	Control Plane and User Plane Latency
User Throughput	Peak throughput Throughput at interference limited Cell edge Cell edge coverage throughput Throughput in different coverage conditions (link budget test) Throughput in different interference conditions (Average User Throughput)
Cell capacity	Cell peak throughput Cell average throughput
Spectral efficiency	User and Cell Spectral efficiency
Coverage	Outdoor Single-cell Coverage Outdoor Multi-Cell Continuous Coverage Outdoor to Indoor Coverage Indoor Coverage
Mobility	Intra-Cell mobility testing Inter-Cell Mobility (Handover) Testing
Reliability	Success probability of transmitting a set number of bytes within a certain delay

Test Case focus	KPIs and Test Scenarios
Retainability	Capability of the network to retain a service by a UE for a desired duration once the user is connected to the service - Call drop rate, Call setup complete rate, Dual connectivity drop rate
User experience	Examples Content Distribution Streaming Service QoE, Social Networking QoE, High Speed Internet QoE
Energy efficiency	UE energy efficiency Network energy efficiency
Inter-RAT procedures	NSA Inter-RAT Mobility with PS Service SA Inter-RAT Mobility with PS Service
RAN architecture split	CU-DU Separation CP-UP Separation
Service or technology specific requirements	
Location/Positioning service	EPS Scenario (Option 3), 5GS Scenario (NG-RAN)
Fixed wireless access	Average and peak throughput
	Meeting throughput speed tier guarantees
	End-to-end latency performance
	Beam tracking range (horizontal/azimuthal) for base station and CPE/UE antennas
	Quality of Service differentiation scenarios for Voice/video application
	Number of (active) users served per base station
	Geographic coverage area
Voice Services	Robustness to environmental blockage (mmWave)
	EPS Fall-back, Voice Over NR

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6.6 Test Conditions to be considered

As highlighted in the earlier sections, the E2E System Test Cases and KPIs which are defined by 3GPP will be directly applicable for testing and validating the O-RAN Deployment Blueprints, while those defined by NGMN in [10] can be leveraged to a large extent for similar purpose.

However, it is important to take into considerations the following aspects when testing and validating O-RAN Deployment Blueprints

1. Multi-vendors interoperability
2. Functional, Performance including normal and abnormal conditions
3. Radio channel variations and impacts on the overall system performance and Quality of Experience (QoE)
4. Management, Orchestration and Network Automation
5. Additional considerations related to practical deployment aspects

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6.6.1 Multi-vendors interoperability

As the O-RAN Deployment Blueprint can consist of multiple Subsystems from possibly multiple vendors, additional considerations for O-RAN focused system test scenarios, conditions and KPIs will be required to validate and evaluate the performance of multi-vendors interoperability in accordance to the O-RAN Deployment Blueprint definition.

For example if the O-RUs, O-DUs and O-CU-CPs/O-CU-UPs in an O-RAN Deployment Blueprint are provided by different vendors, then it would be important to ensure that the E2E KPIs are evaluated with the appropriate E2E test

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cases and test scenarios which can be used to validate multi-vendors interoperability between these O-RAN network functions.

6.6.2 Functional, Performance including normal and abnormal conditions

Multi-vendors interoperability testing typically includes functional validation for handling normal conditions.

However, similar to how the 3GPP Radio Access Networks are expected to be tested and optimized for Performance and Abnormal conditions handling, operators would be expecting the O-RAN Deployment Blueprints to be tested and optimized in the same manner.

O-RAN Deployment Blueprints consisting of Subsystems from multiple vendors will therefore require Multi-vendors interoperability testing to include testing for Performance and Abnormal conditions handling.

Performance testing can be used to evaluate how well each individual Subsystem, each pair of Subsystem Instances and all the Subsystems in the O-RAN Deployment Blueprint inter-operate to achieve the best in class cost-performance ratio.

This can include quantifying the O-RAN Deployment Blueprint when Subsystems from multiple vendors inter-operate under Performance test conditions through

1. Performing testing using the same set of KPIs, for example examining if the latency figures can be achieved in performance testing conditions.
2. Measuring resources utilization efficiency for example with Radio resource blocks, Open Fronthaul and xHaul transport, O-Cloud computing / networking / storage and power consumption / energy efficiency.

Abnormal testing will in some cases be required to be performed to validate fault management and zero-touch network automation techniques such as self-healing functions.

6.6.3 Radio channel variations and impacts on the overall system performance and QoE

Radio access networks should be tested with close to real-world radio channel conditions, variations of the channel conditions and mobility patterns in the lab environment for a proper evaluation of the overall system performance of the RAN and QoE of the services handled by the RAN.

For example testing key 3GPP RAN functions such as Radio resource, Beam / Cell level mobility management and advanced use cases defined by O-RAN such as Traffic Steering and Massive MIMO Beamforming Optimization will require the test environment to be setup with the appropriate network and slices topology, radio channel conditions, UEs distributions and mobility patterns, which otherwise these functions and use cases cannot be properly tested and evaluated.

Testing O-RAN Deployment Blueprints consisting of Subsystems from multiple vendors under the appropriate radio channel conditions and mobility patterns will be required to validate how well the multi-vendors Subsystems inter-operate under these test conditions.

6.6.4 Management, Orchestration and Network Automation

In general, operators require the Management & Orchestration (MANO) and Network Automation capabilities of their networks to be robustly validated.

In order for operators to be able to deploy O-RAN networks and network slices which can consist of Subsystems from multiple vendors, multi-vendors interoperability testing will be required for validating the management and orchestration functions of all the Subsystems defined in the O-RAN Deployment Blueprints.

O-RAN Alliance has specified a set of prioritized use cases for O-RAN orchestration of virtualized RAN and the interfaces used for management and orchestration, in particular the O1 interface between the service management and orchestration framework and the RAN managed functions and the O2 interface between the service management and orchestration framework and the O-Cloud Infrastructure Management Services/Deployment Management Services that controls resource assignment for Virtualized Network Functions.

Operators will require guidance on the E2E Test Cases and KPIs which can be used to test and evaluate the functionality and performance for each of these orchestration use cases in order to be able to quantify the feasibility and benefits of deploying O-RAN compliant networks.

6.6.5 Additional considerations related to practical deployment aspects

In addition to performing testing based on 3GPP, ITU-T, IEEE specified functionalities, it is important as well to take into considerations the practical deployment aspects which operators will need to consider and therefore tested as part of the E2E Testing framework. These considerations include

1. Performance Evaluation and Monitoring
2. Robustness, Resilience, Reliability, Security
3. Troubleshooting and Root Cause Analysis (RCA)
4. Practical Network Operating Conditions such as fronthaul transport loading and non-zero variable delays
5. FCAPS including fault, configuration, accounting, performance, and security management
6. System Deployment lifecycle (installs, upgrades)
7. Virtualization, interaction with O-Cloud (NFVi), Management and Orchestration (MANO)
8. Evolution towards Zero Touch Automation network management

System Deployment lifecycle (installs, upgrades) involves multiple phases of the System Deployment lifecycle which will include new installs and as well as future field upgrades scenarios such as

1. Performing software or firmware upgrades of the existing installed O-RAN Network Functions (e.g. increased performance, enhanced capabilities such as dynamic spectrum sharing, bug fixes and others)
2. Installing new O-RAN Network Functions which are required to interoperate with existing installed O-RAN Network Functions (e.g. adding new O-RUs from a new vendor to the field operational O-DU currently serving O-RUs from the current vendor, new xApp to be installed onto the current near-RT RIC)
3. Installing new or Replacing existing hardware (or O-Cloud) to support existing installed O-RAN Network Functions (e.g. adding, replacing storage, networking, computing resources)

It is therefore important to define testing templates ensuring that the essential testing is performed for the specific Subsystem Instances, Subsystem Instances Pairs and Open Interfaces and System Deployment Blueprint which are affected by the System Deployment lifecycle scenarios.

6.7 Test Scenarios and KPIs to be considered

A few examples of Test Scenarios and KPIs which are needing to be considered for testing O-RAN Deployment Blueprint consisting of Subsystems from multiple vendors are shared in more details in this section.

1. Mobility management
2. Deployability

6.7.1 Mobility Management

Mobility management test cases may need to consider test scenarios which involve testing multi-vendors interoperability between Subsystems provided from multiple vendors.

Test scenarios can include intra O-RU (inter beams for intra cell and inter cells), inter O-RUs (intra O-DU), inter O-DUs (intra O-CU-CP/O-CU-UP), inter O-CU-CP/O-CU-UPs, Intra vs Inter RATs handovers in Idle and Connected modes.

Figure 6-2 shows an example of a test scenario which can be used to verify the interoperability aspects of mobility management in a multi-vendors O-RAN Deployment Blueprint test setup.

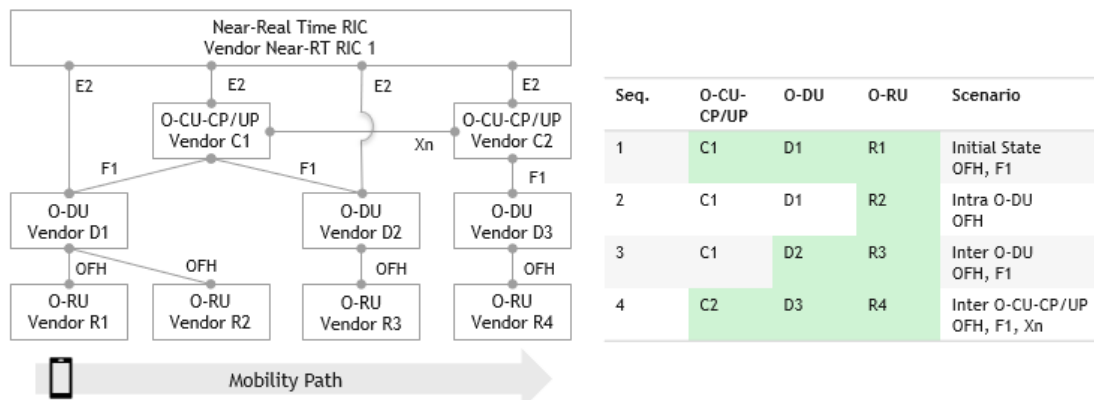


Figure 6-2: Test Scenario for verifying multi-vendors interoperability aspect of Mobility Management

In this test scenario

1. UE is first connected through O-RU (R1), O-DU (D1), O-CU-CP/UP (C1)
2. UE then moves towards O-RU (R2) which is an inter O-RU and intra O-DU mobility scenario
3. UE then moves towards O-RU (R3) which is an inter O-RU and inter O-DU mobility scenario
4. UE then moves towards O-RU (R4) which is an inter O-RU, inter O-DU and O-CU-CP/UP mobility scenario

The Mobility KPIs including handover attempts/success rates, interruption time (aka latency) to control, user plane and services plane, services performances such as packet losses, jitter and latency should be measured in order to validate that the services SLAs can be met while executing the test sequences.

6.7.2 Deployability

One of the key considerations for dis-aggregated RAN deployments is the requirements for the underlying transport network which will be dependent on the requirements of the Services and dis-aggregated RAN functions which will be required to interwork over the transport network segments.

These requirements are typically simulated, estimated, tested, measured, and validated for dis-aggregated RAN transport planning purposes.

O-RAN Deployment Blueprints will be required to go through similar process with the additional complexity that the Subsystems in the same O-RAN Deployment Blueprint may be provided by different vendors which may have different levels of influences on the transport requirements therefore needing to be tested and validated with test setup considerations outlined in Section 6.6.

The Integrity and Utilization KPIs including transport capacity/throughput, packet losses, jitter and latency should be measured during the testing process taking into considerations the following aspects

1. Capacity/throughput for Open fronthaul interface for transport planning purposes, which can vary depending on test scenarios and key factors such as radio conditions among others outlined in Section 6.6.3.
2. Latency for transport planning purposes and delay management interoperability which can vary dependent on the transport options and can be impacting performance and users-services experiences.
3. Resources Utilization based on test profiles which should include networking, computing, storage resources.

Annex A: Operators' inputs on Deployment Blueprints

Annex A lists the O-RAN Deployment Blueprint profiles which have been submitted by global operators to O-RAN Alliance to assist with developing comprehensive E2E testing methodologies and test cases taking into considerations these submitted profiles.

A.1 Operator #1 inputs

NR Deployment Options and Evolution

1. Step1: (NSA EN-DC) Option 3x (based on X2 interface and EPC)
2. Step2: Step1 + (SA 5G) Option 2
3. Step3: Step2 + (ng-LTE) Option 5/Option 7x (based on Xn interface and 5GC)

Use Cases focuses

1. RAN Sharing
2. Near-RT RIC (edge-cloud) Non-RT-RIC Use Cases (regional-cloud): Traffic Steering, QoE/QoS, Massive MIMO Beam Forming and Optimization

E2E Performance requirements of Application / Network Slices – references to NGMN Definition of the Testing Framework for the NGNM 5G Pre-Commercial Network Trials [10].

Table A-1: Operator #1 inputs - System Profile considerations

System Profiles considerations	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Service Priority	eMBB (NR and LTE)	eMBB (NR and LTE)	eMBB (LTE)	eMBB (NR), URLLC (NR)	eMBB (NR), URLLC (NR)
Deployment Scenarios	Outdoor Macro/Micro Cell and RAN Sharing				
Deployment Options and Evolution	NSA EN-DC Option 3x with X2 between eNB and en-gNB; and EPC	NSA Option 7x with Xn between gNB and ng-eNB; and 5GC	(ng-LTE) SAS Option 5 with 5GC	NR Standalone (SA) Option 2 with 5GC	NR Standalone (SA) Option 2 with 5GC
NR Spectrum Bands	NR: FR1 - 3.7 GHz TDD, 700 MHz FDD LTE: 2.6GHz/1800MHz/800MHz FDD	NR: FR1 - 3.7 GHz TDD, 700 MHz FDD LTE: 2.6GHz/1800MHz/800MHz FDD	2.6GHz/1800MHz/800MHz FDD	FR1 - 3.7 GHz TDD, 700 MHz FDD	FR2 - 26 GHz TDD
Figure Reference	Figure A-1	Figure A-1	Figure A-2	Figure A-2	Figure A-3
Reference to WG6 CAD [4]	Based on Scenario A in Appendix 7.1	Based on Scenario A in Appendix 7.1	Based on Scenario A in Section 6.1	Based on Scenario A in Section 6.1	Based on Scenario E in Section 6.1

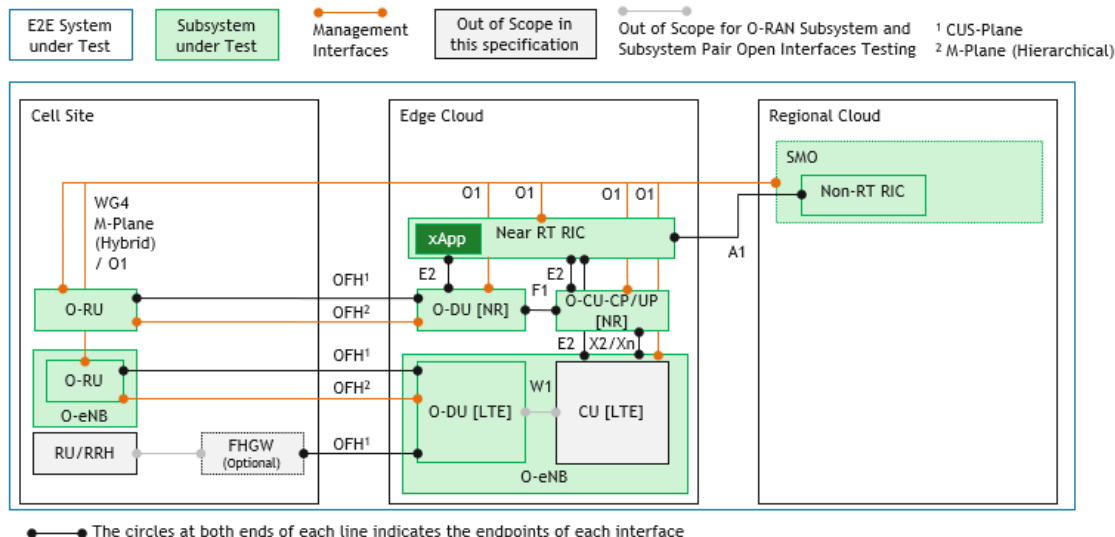


Figure A-1: Operator #1 Deployment Scenario #1 – NSA (FR1)

Note: W1 is specified in 3GPP Release 16 for ng-eNB only.

Refer to [4][11][12] for more details on the FHGW.

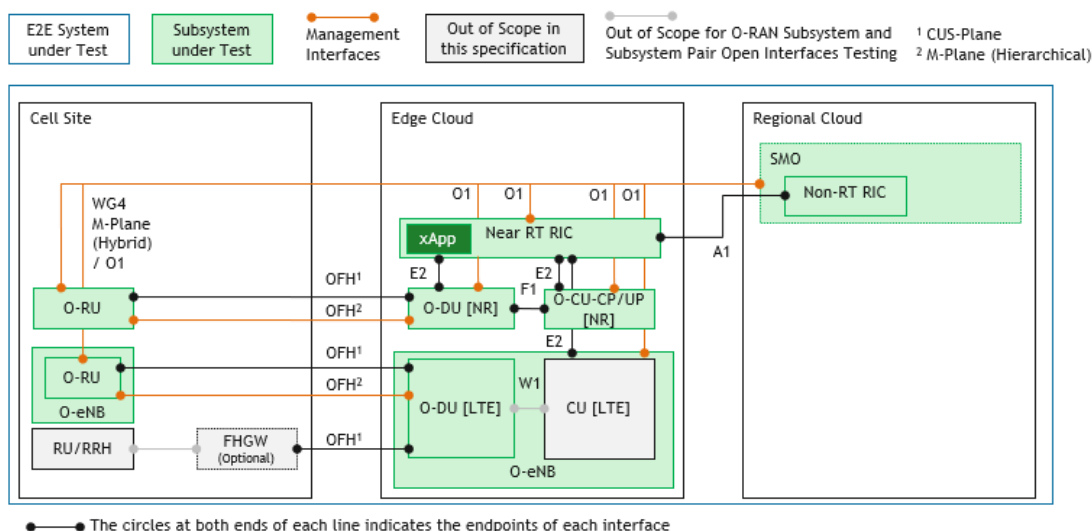


Figure A-2: Operator #1 Deployment Scenario #2 – SA (FR1)

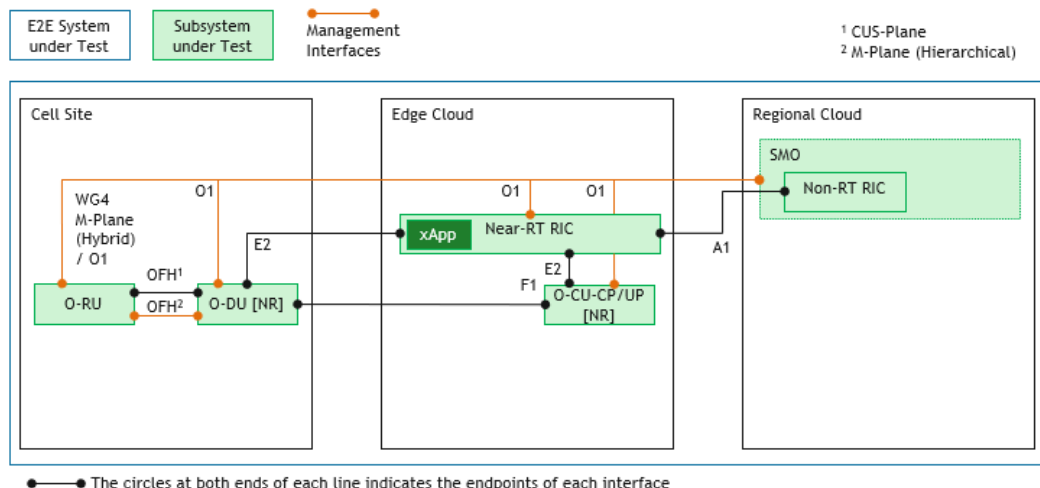


Figure A-3: Operator #1 Deployment Scenario #3 – SA (FR2)

Table A-2: Operator #1 inputs – Subsystem O-RU

O-RU	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Separated / Integrated O-DU and O-RU	Separated: WG4: OFH				Integrated or Separated. In the case if the O-DU and O-RU are integrated, OFH is not available for testing. In the case if the O-DU and O-RU are implemented as separate nodes and they can either be co-located in the same or distributed locations – WG4: OFH.
Management Plane (WG4 M Plane Hierarchical, Hybrid mode, O1)	WG4: M-Plane specification; WG1: O1				
PNF, VNF, CNF	PNF				
Location	Cell Site				

Table A-3: Operator #1 inputs – Subsystem FHGW

FHGW	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
7-2x <-> 7-2x; 7-2x <-> 8	7-2x <-> 7-2x; 7-2x <-> 8 (LTE)	-	-	7-2x <-> 7-2x	7-2x <-> 7-2x (if DU-RU not integrated)

Table A-4: Operator #1 inputs – Subsystem O-DU

O-DU	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Separated / Integrated O-DU and O-RU	Separated WG4: OFH				Integrated or Separated. In the case if the O-DU and O-RU are integrated, OFH is not available for testing. In the case if the O-DU and O-RU are implemented as separate nodes and they can either be co-located in the same or distributed locations – WG4: OFH.

O-DU	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Separated / Integrated O-DU and O-CU-CP/O-CU-UP	Separated WG5: F1, W1	Separated WG5: F1, W1	Separated WG5: W1	Separated WG5: F1	Separated WG5: F1
Management Plane (WG4 M Plane Hierarchical, Hybrid mode, O1)	WG4: M-Plane specification; WG1: O1				
PNF, VNF, CNF	WG6: VNF, CNF				PNF
O-Cloud	WG6: O2				N/A
Location (Cell, Edge site)	WG6: Edge				Cell
Near-Real time RIC support (counters and stats perspective, E2)	WG3: E2				

Table A-5: Operator #1 inputs – Subsystem O-CU-CP/ O-CU-UP

O-CU-CP /O-CU-UP	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Management	WG1: O1				
Separated / Integrated / O-DU and O-CU-CP/O-CU-UP	Separated WG5: F1, W1	Separated WG5: F1, W1	Separated WG5: W1	Separated WG5: F1	Separated WG5: F1
PNF, VNF, CNF	WG6: VNF, CNF				
O-Cloud	WG6: O2				
NR Deployment Options (NSA Option 3x, SA Option 2, others)	WG5: X2 (NSA)	WG5: Xn (SA)			
Location (Edge, Regional site)	WG6: Edge				
Near-Real time RIC support (counters and stats perspective, E2)	WG3: E2				

Table A-6: Operator #1 inputs – Subsystem O-eNB

O-eNB	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Management	WG1: O1			N/A	N/A
PNF, VNF, CNF	WG6: VNF, CNF			N/A	N/A
O-Cloud	WG6: O2			N/A	N/A

O-eNB	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
NR Deployment Options (NSA Option 3x, SA Option 2, others)	WG5: Xn (SA), X2 (NSA)		WG5: Xn (SA)	N/A	N/A
Location (Cell, Edge, Regional site)	WG6: Edge			N/A	N/A
Near-Real time RIC support (counters and stats perspective, E2)	WG3: E2			N/A	N/A

Table A-7: Operator #1 inputs – Subsystem O-Cloud

O-Cloud	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Management	WG6: O2				
Location (Cell, Edge, Regional site)	WG6: Edge, Regional				

Table A-8: Operator #1 inputs – Subsystem Near-Real Time RIC

Near-Real Time RIC	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Base functions	WG3: E2, xApp; WG2: A1				
PNF, VNF, CNF	WG6: VNF, CNF				
O-Cloud	WG6: O2				
Location (Edge, Regional site)	WG6: Edge				
xApps	Traffic Steering, QoE/QoS				
Advanced use cases (such as Network Slicing)	Massive MIMO Beamforming Optimization (WG1), RAN sharing (WG1)				

Table A-9: Operator #1 inputs – Subsystem Non-Real Time RIC

Non-Real Time RIC	NSA Opt. 3x (FR1)	NSA Opt. 7x (FR1)	SA Opt. 5 (FR1)	SA Opt. 2 (FR1)	SA Opt. 2 (FR2)
Base functions	WG2: A1				
PNF, VNF, CNF	WG6: VNF, CNF				
Location (Edge, Regional site)	WG6: Regional				

Table A-10: Operator #1 inputs – Subsystem SMO

Table A-11: Operator #1 inputs – Subsystem xApp

A.2 Operator #2 inputs

1. Indoors Small Cell
2. 5G NR FR1 2.6GHz, 4.9GHz – TDD
3. Maximum Channel Bandwidth (CHBW) 100MHz
4. Leverages WG4 Shared Cell feature with combination of both FHM and Cascade modes

1. NR NSA Option 3x
2. NR SA Option 2

1. Phase 1: O-RU, FHGW, O-DU/O-CU-CP/O-CU-UP (integrated unit) and Server
2. Phase 2: O-RU, FHGW, O-DU/O-CU-CP/O-CU-UP (integrated unit), AAL and Server

Refer to [4][11][12] for more details on the FHGW.



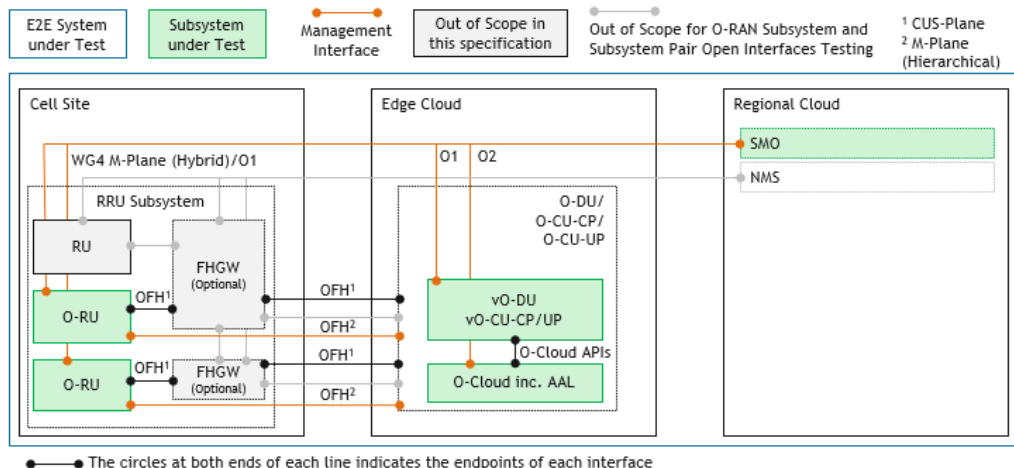


Figure A-5: Operator #2 Deployment Scenario #1 (FR1) – Phase 2

A.3 Operator #3 inputs

Deployment Scenario

1. Outdoor Macro Cell Deployment
2. 5G NR FR1 n66, n70, n71, n29, n26
3. Total Channel Bandwidth (CHBW) 150MHz

Deployment Options and Evolution

1. NR SA Option 2

O-RAN Deployment Blueprints variants and corresponding Subsystems

1. Refer to Figure A-6: O-RU, O-DU at Cell site, O-CU-CP, and O-CU-UP at Regional cloud, AAL and Server
2. Refer to Figure A-7: O-RU, O-DU at Local cloud, O-CU-CP, and O-CU-UP at Regional cloud, AAL and Server

Use Cases focuses

1. White Box Hardware and Open Interfaces
2. Near-RT RIC (edge-cloud) and Non-RT-RIC Use Cases (Regional cloud): Traffic Steering, QoS /QoE Optimization and Slicing
3. Cloud Deployment

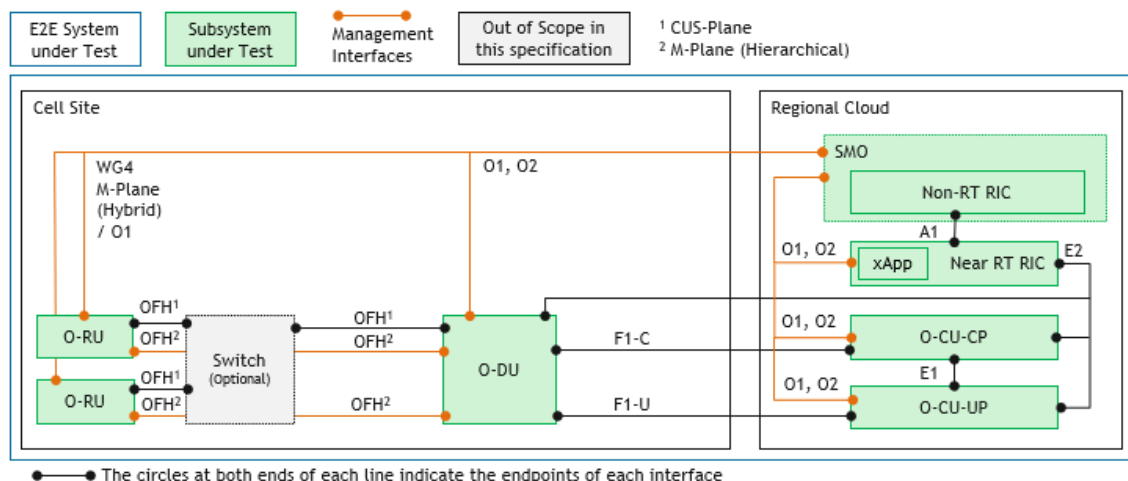


Figure A-6: Operator #3 Deployment Scenario #1 – SA (FR1)

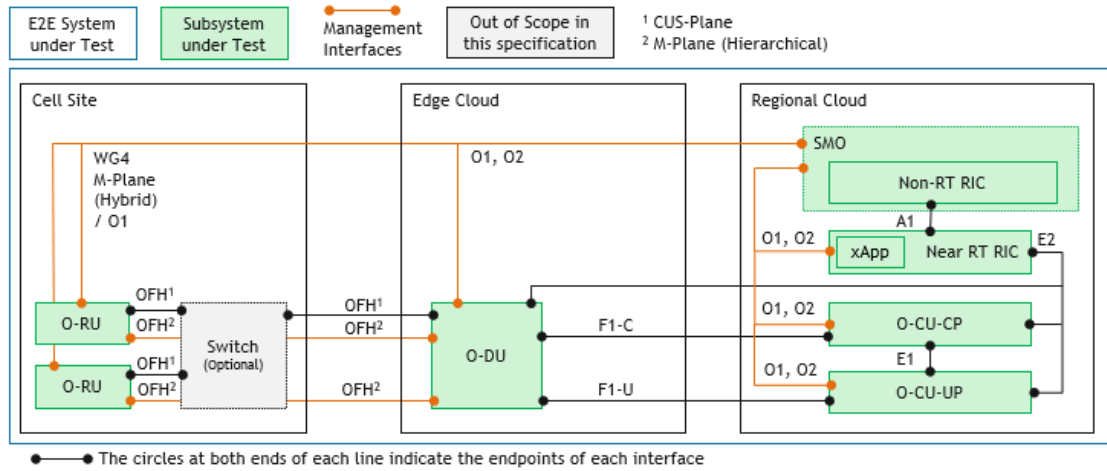


Figure A-7: Operator #3 Deployment Scenario #2 – SA (FR1)

A.4 Operator #4 inputs

Deployment Scenario

1. Outdoor Small Cell, Outdoor Macro Cell, Indoors Small Cell
2. 5G NR FR1 TDD (3.7GHz, 4.5GHz), 5G NR FR2 TDD (28GHz)
3. Maximum Channel Bandwidth (CHBW) 100MHz for FR1 and 400MHz for FR2

Deployment Blueprints variants and corresponding Subsystems

Table A-12: Operator #4 inputs – Deployment Blueprint variants

Deployment variants	NSA Opt. 3x (NSA1)	NSA Opt. 3x (NSA2)	SA Opt. 2 (SA1)	NSA Opt. 3x (NSA3)	NSA Opt. 3x (NSA4)	SA Opt. 2 (SA2)
Subsystems	O-RU, Integrated O-DU/O-CU-CP/O-CU-UP	O-RU, Integrated O-DU/O-CU-CP/O-CU-UP, (Shared Cell) FHM	O-RU, Integrated O-DU/O-CU-CP/O-CU-UP	Integrated O-RU/O-DU, O-CU-CP/O-CU-UP	O-RU, Integrated O-DU/O-CU-CP/O-CU-UP, Near-RT RIC, xApps, SMO	O-RU, Integrated O-DU/O-CU-CP/O-CU-UP, O-Cloud, SMO
Interfaces under Test	X2, OFH, WG4 M-Plane Hierarchical	X2, OFH, WG4 M-Plane Hierarchical	OFH, WG4 M-Plane Hierarchical	X2, F1	X2, OFH, WG4 M-Plane Hierarchical, A1 E2 is an internal interface	OFH, WG4 M-Plane Hierarchical, O1, O2, O-Cloud APIs
PNF, VNF	All PNFs	All PNFs	All PNFs	All PNFs	All PNFs	Virtualized Integrated O-DU/O-CU-CP/O-CU-UP Rest are PNFs
Figure Reference	Figure A-8	Figure A-8	Figure A-9	Figure A-10	Figure A-11	Figure A-12
Reference to WG6 CAD [4]	Based on Scenario A in Appendix 7.1	Based on Scenario A in Appendix 7.1	Based on Scenario A/B in Section 6.1	Based on Scenario A/B in Appendix 7.1	Based on Scenario A in Appendix 7.1	Based on Scenario A/B in Section 6.1

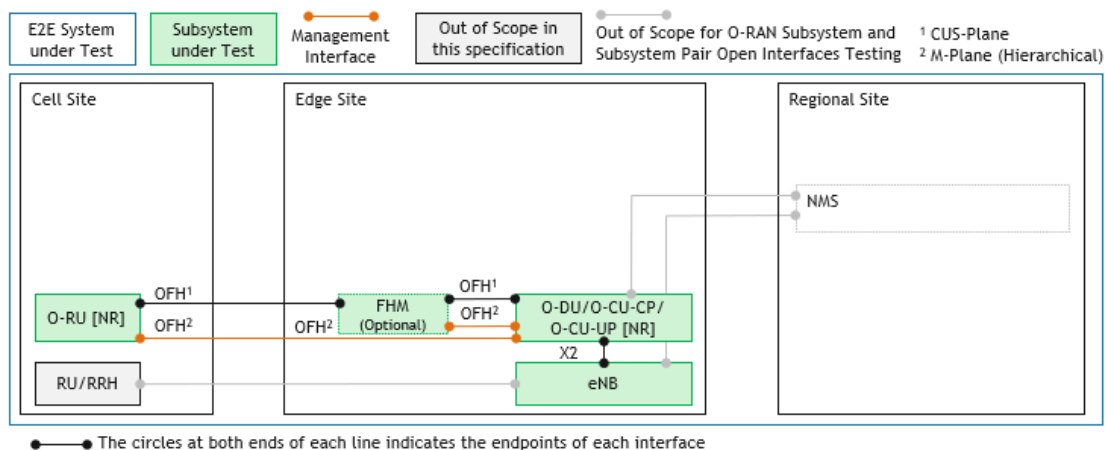


Figure A-8: Operator #4 Deployment Scenario #1 – NSA

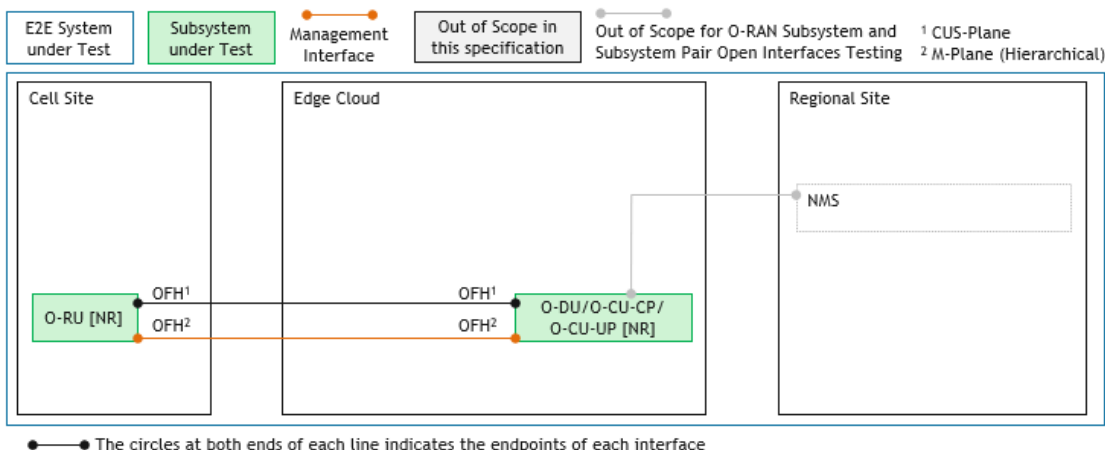


Figure A-9: Operator #4 Deployment Scenario #2 – SA

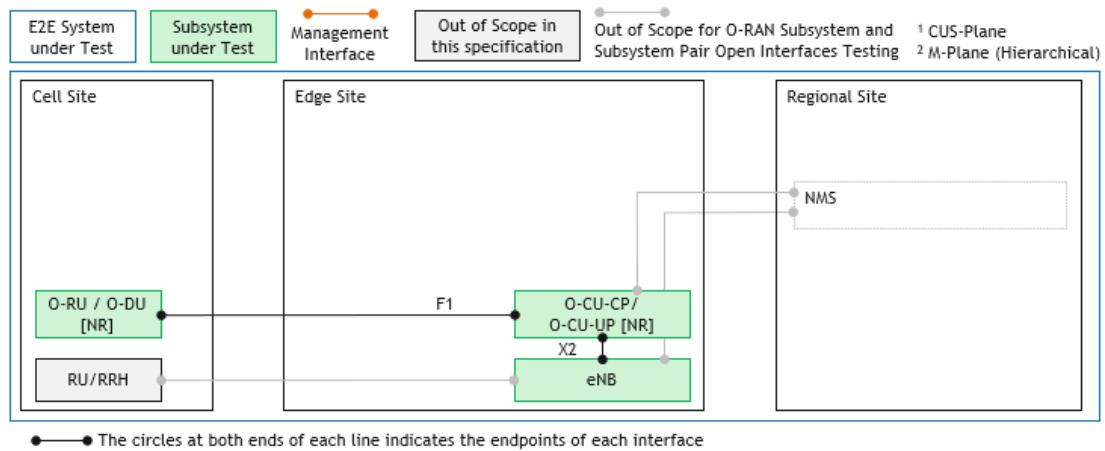


Figure A-10: Operator #4 Deployment Scenario #3 – NSA

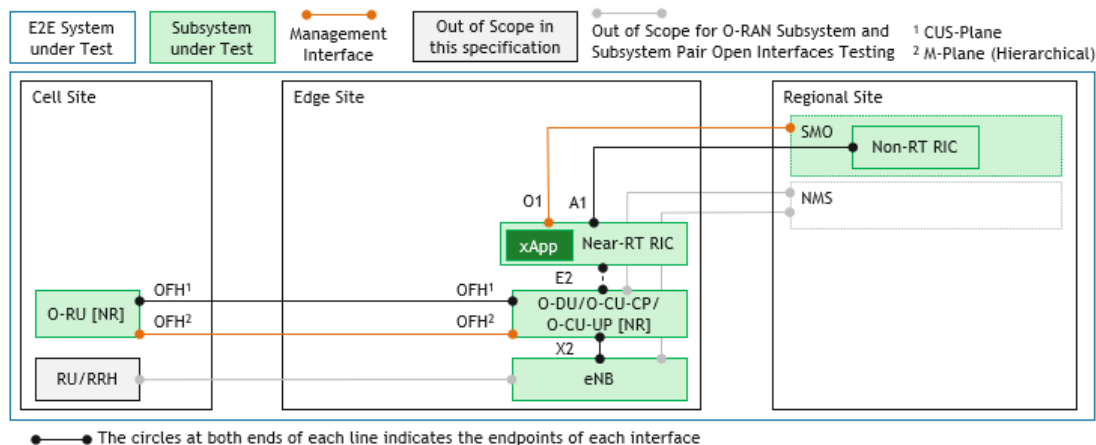


Figure A-11: Operator #4 Deployment Scenario #4 – NSA

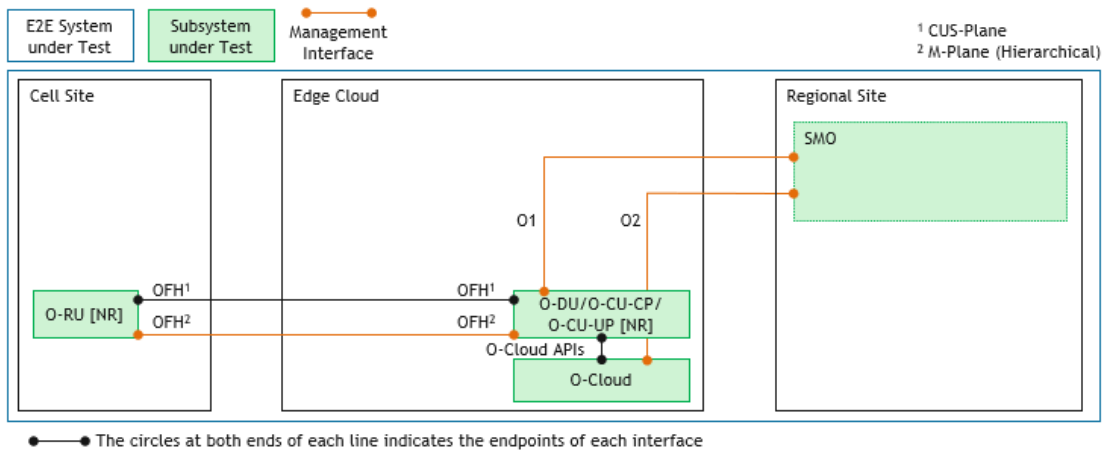


Figure A-12: Operator #4 Deployment Scenario #5 – SA

Annex B: O-RAN Software Community inputs on Testing Methodology

This annex provides an overview on the O-RAN Software Community (OSC) work and its current software releases and testing practices. Refer to the OSC wiki site for the most current information (<https://wiki.o-ran-sc.org/>).

The O-CU referred in this annex should be considered as O-CU CP/UP.

B.1 O-RAN Software Community

The O-RAN Software Community (OSC) is an open source software development project aimed at providing an open source implementation of an end-to-end RAN system following O-RAN Alliance architecture and specifications. The OSC is hosted by the Linux Foundation and funded by the O-RAN Alliance.

The OSC is led by a 12-member Technical Oversight Committee (TOC) with its current members representing AT&T, China Mobile, Deutsche Telekom, Ericsson, Nokia, NTT DOCOMO, Orange, Radisys and Verizon.

The working effort of the OSC is organized into the following development projects:

- Operations and Maintenance (OAM)
- Non-Realtime RAN Intelligent Controller (NONRTRIC)
- Near-Realtime RAN Intelligent Controller Platform (RIC)
- Near-Realtime RAN Intelligent Controller Applications (RICAPP)
- O-RAN Central Unit (OCU)
- O-RAN Distributed Unit High (ODUHIGH)
- O-RAN Distributed Unit LOW (ODULOW)
- Simulations (SIM)
- Infrastructure (INF)
- Documentation (DOC)
- Integration and Testing (INT)

and subcommittee(s)

- Requirements and Software Architecture Committee (RSAC)

The OSC conducts its business on open communication and development platforms including Zoom teleconferencing bridge, Groups.io mailing lists, Confluence wiki, and Git/Gerrit source code repositories. All O-RAN SC information and O-RAN SC developed software can be accessed via resources listed in Annex B.6.

B.2 OSC Development Cycles

The OSC developed software is released on a bi-annual schedule, first in May/June time frame and the second in November/December. The naming scheme of releases follows color names in alphabetical order, with individual release names selected by community member voting. The first four releases are named Amber, Bronze, Cherry, and Dawn. The OSC delivered its first release, Amber release, in November 2019.

The high-level objectives and use cases for the next release are usually selected by the RSAC during the later portion of the current release cycle. The first several weeks of each release cycle typically are for the design and planning phase where the high-level release objectives are designed into implementable features and tasks, which are distributed to the OSC development projects. Following the planning phase is the development phase, during which the development projects and teams implement the agreed-upon features. This phase is divided into three-week sprints where each sprint has its own more manageable implementation and testing goals. Finally, prior to the release of the software, it is the testing phase, during which the integrated system testing and end-to-end use case testing are conducted, and the viability and quality of the software are verified and evaluated.

After each release, the OSC also institutes a maintenance window for the new release during the design and planning phase of the next release cycle. Normally this time window is used only for fixing major problems discovered after the release.

B.3 OSC Software and Test Deployment Architecture

The OSC projects are set up based on the O-RAN Alliance architecture.

Figure B-1 below shows the relationship between OSC projects and the O-RAN architecture components.

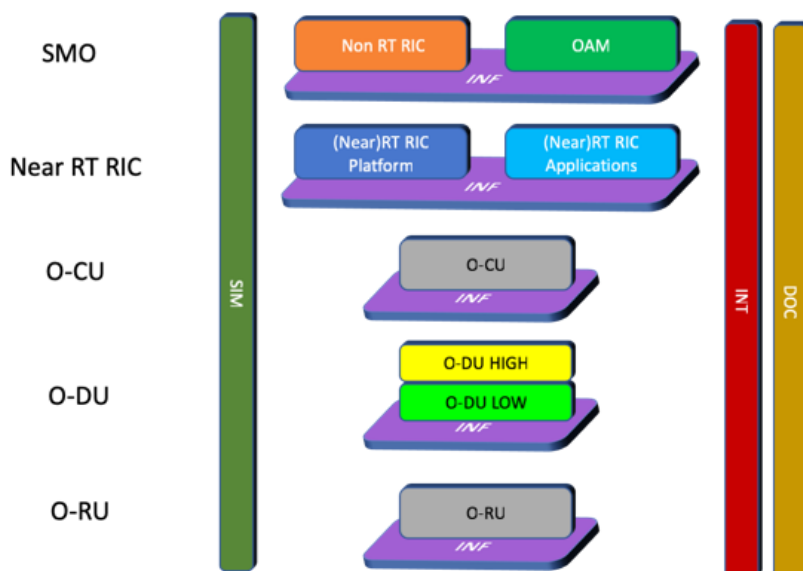


Figure B-1: Relationship between OSC projects and the O-RAN architecture components

Figure B-2 depicts how the delivered software components by the OSC projects are deployed in the OSC Integration and Testing lab for Bronze release testing.

The deployment is spread across two Kubernetes clusters and two bare metal servers. From right to left, the first Kubernetes cluster is the SMO cluster, for components developed by the Non-Realtime RIC and OAM projects. The second Kubernetes cluster is the RIC cluster, for deploying components and microservices developed by the Near Realtime RIC Platform and the Near-Realtime RIC Applications projects. One of the bare-metal servers is used for running O-DU (both O-DO LOW and O-DU HIGH), and the other for running O-RU stub for peering the Open Fronthaul connection to the O-DU, over a dedicated 40GE link.

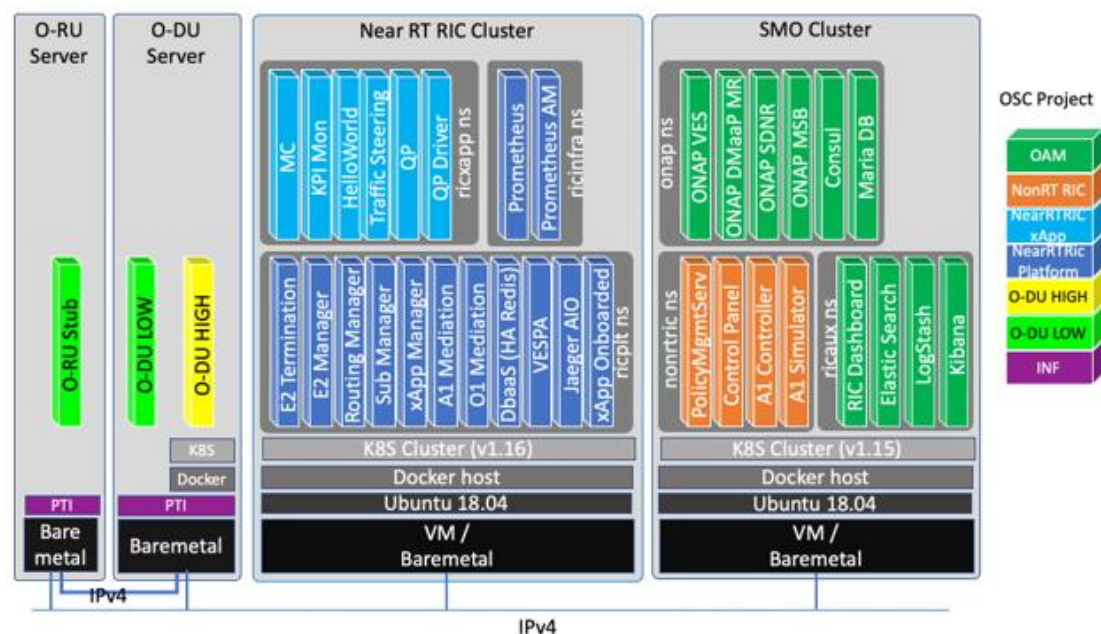


Figure B-2: Deployment of OSC projects components in the OSC Integration and Testing lab for Bronze release

B.4 OSC Release Objectives

Amber (achieved)

- O-RAN SC level
 - Continuous Integration (CI) flow implemented and all source code repositories integrated into the CI auto building process.
 - Source code repositories build into binary artifacts (i.e. docker container images, libraries, OS ISO media).
 - Documentation framework established, initial documentation in place.
- Selected projects
 - Near-Real Time RIC Platform and Near-Real Time RIC Applications:
 - “one-click” deployable.
 - Completing xApp deployment flow, (pre-spec) E2 setup and subscription flow.

Bronze (work in progress)

- O-RAN SC level
 - Unit Test and SONAR reporting framework established.
 - All participating projects “one-click” deployable.
 - RSAC use cases:
 - Health-check: including A1 and (pre-spec) O1 health check.
 - Traffic Steering: steering decision based on E2 information and AI/ML algorithm.
 - O-DU integration: O-DU HIGH communicate with O-DU LOW via FAPI.

B.5 OSC Testing

OSC testing effort is divided into five levels

1. Unit testing (UT)

- UT is performed by the developers by implementing test cases using programming language specific unit testing and static code analysis framework.

- UT is triggered automatically every time when there is source code change submitted into the code repository for review.
 - At OSC level the INT project team provides support for unit test and code scan (i.e. static code analysis) integration into Linux Foundation Continuous Integration flow, and result reporting into the SONAR Cloud platform. Using the reports, TOC sets quality gates such as percentage of code covered by unit test cases, no major issues discovered, etc.
2. Project level integrated testing (PIT)
 - Each development project is expected to conduct project level integrated testing, where all software components developed by the project are deployed into a single system for testing internal flows.
 - PIT is expected to be conducted by individual project teams, at the end of each development sprints or other development milestones.
 - At OSC level the INT project team organizes sprint demonstration fests regularly where development project teams gather and demonstrate achieved features in the previous sprint, using project level system deployment in project team's own testing and demonstration environment.
 - The INT project team is also working with individual development projects on Robot testing framework-based testing flows that are to be integrated into Linux Foundation Continuous Integration flow. Such test flows serve as regression tests for implementation updates and can be triggered automatically by the CI system on a regular basis (e.g. daily) or upon new code submission.
 3. Project pairwise testing (PPT)
 - For projects that deliver O-RAN architectural components that interact with each other, PPT is conducted to ensure communication compatibility.
 - PPT sessions are organized by the INT project team, using either community testing lab or self-arrange testing facilities by involved projects.
 4. System integrated testing (SIT)
 - SIT is whole OSC software testing where all OSC development components are deployed into a single system. SIT focuses on overall system health, inter-project communication and application compatibility, overall system deployment flow, and resource requirements.
 - SIT is conducted by the INT project team using OSC community testing lab.
 - SIT is conducted during the testing phase of each release cycle.
 - Robot test framework-based health check test flows used as automated regression test.
 5. Use case testing (UCT)
 - UCT is whole OSC software testing where all OSC development components are deployed into a single system. UCT tests OSC deployment that has passed SIT.
 - UCT is conducted by the INT project team using OSC community testing lab.
 - UCT is conducted during the testing phase of each release cycle.
 - UCT uses Open Testing Framework (OTF), a test flow definition and orchestration tool that is developed as a sub-project under the INT.

B.6 References

1. OSC Resources
 - a) Wiki: <http://wiki.o-ran-sc.org>
 - b) Source code: <http://gerrit.o-ran-sc.org>
 - c) Project management: <http://jira.o-ran-sc.org>
 - d) Docker image registry: nexus3.o-ran-sc.org:10004 and nexus3.o-ran-sc.org:10002
 - e) SONAR reporting: <https://www.sonarcloud.io/organizations/o-ran-sc/projects>
 - f) Mailing lists: <https://lists.o-ran-sc.org/g/main>
 - g) Meeting calendar: <https://lists.o-ran-sc.org/calendar> (member sign-in required)
2. Robot Framework: <https://robotframework.org>

Annex C: Test Functions, Tools and Solutions for Subsystem and E2E System Testing

Table C-1 shows an example listing of the test functions, tools and solutions which can be used to establish test bed setups for Subsystem testing, Subsystem Pairing and Open Interfaces Interoperability testing and E2E System testing of the O-RAN Deployment Blueprints.

These test functions and test tools are classified into the following categories as they can be used for setting up the test environment, providing active stimulus functions and passive measurements for test results verification purposes.

- Test Case Scenario Creation:** used to setup the testing environment and the necessary testing conditions which are required for executing the test case.
- Test Case Validation:** used to collect measurements, KPIs, logs, packet captures for verifying, substantiating, troubleshooting, and debugging test results.

A single test function and tool can be used for both Test Case Scenario Creation and Test Case Validation.

Few notes for clarification purposes on the example listings of test functions and tools listed in this Annex C

- These are non-exhaustive listings i.e. test functions and tools which are not listed in this Annex C can be used where appropriate.
- Not all the test functions and tools listed will be required for all test scenarios and test cases.
- Test functions and tools listed can be implemented each as a standalone test tool or multiple of these test functions can be combined into a single test tool.
- Few of the test functions might be combined into a single test tool (standalone test function usage is not possible).
- Real or emulated network elements / network functions can be used where appropriate.

This example listing will be expanded in the next release of this specification.

Table C-1: Subsystem and System Testing – Example listing of Test Functions and Tools

No.	Test Functions and Tools	Abbreviation	Test Case Scenario Creation	Test Case Validation	Purpose	Subsystem and Subsystem Pairing and Open Interfaces Interoperability Testing	E2E System Test
1	Test UEs with Test SIMs (LTE, NR NSA, NR SA)	Test UE	Yes	Yes	Used for test cases which require UEs interactions with the SUT. Test UEs are typically UEs which are designed for commercial or testing applications with certain test and diagnostic functions enabled for test and measurements purposes.	Yes	Yes
2	Multi UEs Emulator (LTE, NR NSA, NR SA)	M-UE Emu	Yes	Yes	Used for test cases which require UEs interactions with the SUT.	Yes	Yes
3	Test Core Network	Test Core	Yes	Yes	Used to terminate Test UEs and/or UEs emulator NAS	Yes	Yes

No.	Test Functions and Tools	Abbreviation	Test Case Scenario Creation	Test Case Validation	Purpose	Subsystem and Subsystem Pairing and Open Interfaces Interoperability Testing	E2E System Test
	(EPC, NSA EPC, 5GC)				protocol and to support core network procedures required for RAN (SUT) testing. Test Core Network are typically Core Networks which are designed for commercial applications and can be used for testing the RAN.		
4	Core Network Emulator (EPC, NSA EPC, 5GC)	Core Emu	Yes	Yes	Used to terminate Test UEs and/or UEs emulator NAS protocol and to support core network procedures required for RAN (SUT) testing.	Yes	Yes
5	Test eNB or gNB (NSA, SA)	Test eNB or Test gNB	Yes	Yes	Used for test cases which require eNB or gNB interaction with DUT/SUT.	Yes	Yes
6	eNB or gNB Emulator (NSA, SA)	eNB Emu or gNB Emu	Yes	Yes	Used for test cases which require eNB or gNB interaction with DUT/SUT.	Yes	Yes
7	O-RU Emulator testing the O-DU	O-RU OFH Emu	Yes	Yes	Used for test cases which require O-DU Subsystem testing.	Yes	No
8	O-DU Emulator testing the O-RU	O-DU OFH Emu	Yes	Yes	Used for test cases which require O-RU Subsystem testing.	Yes	No
9	O-DU Emulator testing the O-CU-CP/O-CU-UP	O-DU F1 Emu	Yes	Yes	Used for test cases which require O-CU-CP/O-CU-UP Subsystem testing.	Yes	No
9	O-CU-CP/O-CU-UP Emulator testing the O-DU	O-CU-CP/O-CU-UP F1-C/F1-U Emu	Yes	Yes	Used for test cases which require O-DU Subsystem testing (in the case of O-DU is implemented as a standalone O-DU).	Yes	No
10	E2 Node Emulator (O-CU-CP/O-CU-UP, O-DU)	O-CU-CP/O-CU-UP E2 Emu or O-DU E2 Emu	Yes	Yes	Used for test cases which require Near-Real Time RIC and xApps Subsystems testing.	Yes	No
11	Application Test Server (aka Traffic Generator)	App Test Svr or Traffic Gen	Yes	Yes	Used for test cases which require bi-directional (downlink and uplink) user plane	Yes	Yes

No.	Test Functions and Tools	Abbreviation	Test Case Scenario Creation	Test Case Validation	Purpose	Subsystem and Subsystem Pairing and Open Interfaces Interoperability Testing	E2E System Test
					transfer test(s) to be performed between the Application Test Server and Test UEs and/or UEs emulator.		
12	Network Impairment Emulator	NW Impair Gen	Yes	No	Used for test cases which require selective packet impairments such as (but not limited to) over the Open Fronthaul (OFH) interface, X2 and others as appropriate.	Yes	Yes
13	Packet Delay Insertion Generator	Pkt Delay Gen	Yes	No	Used for test cases which require selective insertion of packet delay such as (but not limited to) over the Open Fronthaul (OFH) interface and others as appropriate. Noting that fixed delay can be managed via a specific length of fiber as well as using dedicated test tool.	Yes	Yes
14	Transport and Synchronization Tester	Transport Sync Tester	Yes	Yes	Used for test cases which require passive and active testing of transport network across the fronthaul, midhaul, backhaul and sidehaul (Xn, X2) - S-Plane timing and sync verification across all O-RAN components in both subsystem and system test cases. Supports eCPRI testing for O-RU, O-DU, FHM and FTN.	Yes	Yes
15	RF Attenuators	RF Atten.	Yes	No	Used for test cases which require radio signals attenuation and can also improve impedance match, ensuring power transfer from source to load is optimized.	Yes	Yes

No.	Test Functions and Tools	Abbreviation	Test Case Scenario Creation	Test Case Validation	Purpose	Subsystem and Subsystem Pairing and Open Interfaces Interoperability Testing	E2E System Test
16	Channel Emulator	Channel Emu.	Yes	No	Used for test cases which require radio channel emulation / impairments.	Yes	Yes
17	RF Chamber	RF Chamber	Yes	No	Used for Over the Air (OTA) RF connectivity to the O-RU (where applicable).	Yes	Yes
18	Spectrum and Signal Analyzer	SpecAn or SigAnalyzer	No	Yes	Used for test cases which require radio signal and spectrum analysis.	Yes	Yes
19	Signal Creation Software	SCS	Yes	Yes	Used to create waveforms to stimulate NR and LTE test signals.	Yes	No
20	Signal Generator	SigGen or Sig Generator	Yes	No	Used for test cases which require radio signal generation.	Yes	No
21	Test UE Logging Tool	Test UE Logger	No	Yes	Used for test results and KPIs reporting when Test UEs is/are used.	Yes	Yes
22	Protocol Analyzer	Protocol Analyzer	No	Yes	Used for passive monitoring of the Interface under Test (IUT) for test results verification and troubleshooting purposes such as (but not limited to) over the Open Fronthaul (OFH) interface, X2 and others as appropriate. Test Tool can be split into Packet Capture and Protocol Decoder functions. These capabilities can be implemented as a single integrated test tool or separate test tools.	Yes	Yes
23	IQ Analyzer	IQA	No	Yes	Used to analyze IQ data on the Fronthaul interface.	Yes	Yes
24	Network Synchronization Timing Source	NW Sync Timing Source	Yes	No	Used as network timing synchronization source such as the Precision-Time-	Yes	Yes

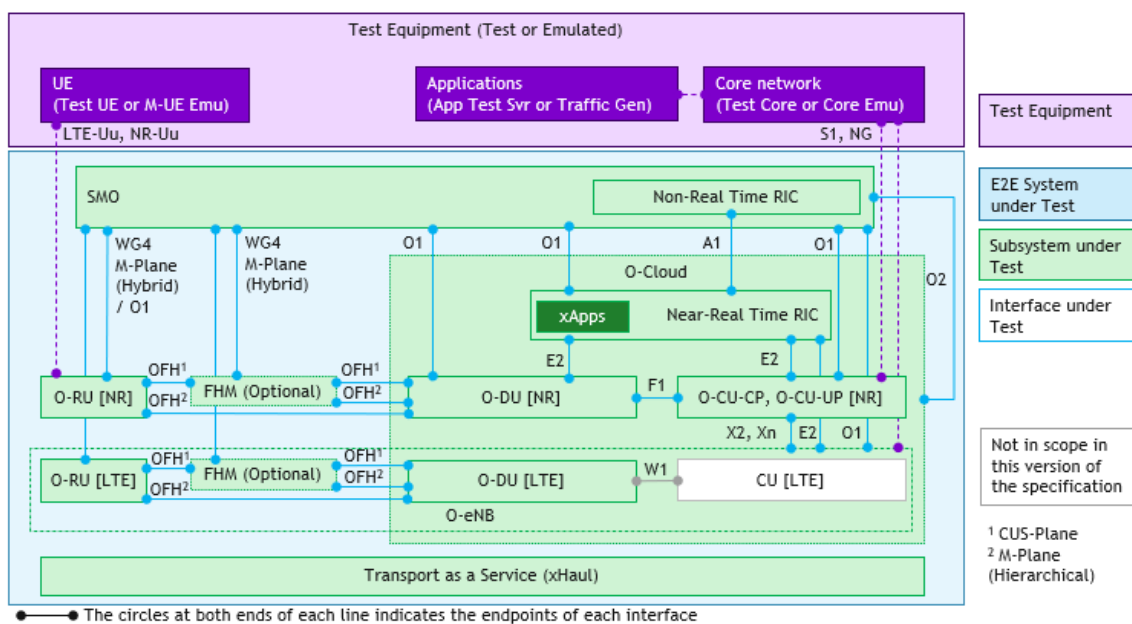
No.	Test Functions and Tools	Abbreviation	Test Case Scenario Creation	Test Case Validation	Purpose	Subsystem and Subsystem Pairing and Open Interfaces Interoperability Testing	E2E System Test
					Protocol (PTP) Grand Master.		
25	Test Controller / Automation	Test Controller and Automation	Yes	Yes	Used for testing orchestration and automation which includes test cases selection, scheduling, configuration of the DUT, SUT, Test functions and Tools in accordance to the selected test cases and test conditions. Test tool steps through the procedures of the test cases based on the current status and conditions of the test cases execution. Test results are analyzed and reported after the execution of the test cases.	Yes	Yes

1

Figure C-1 shows an example with the wrap around testing setup for E2E System under Test (SUT) with the Test functions and tools abbreviations updated.

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3



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Figure C-1: Wrap around testing setup for E2E System under Test (SUT) with Test Functions/Tools Abbreviations updated

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Annex D: References to O-RAN WG6 Cloud Deployment Scenarios

Annex D shows additional Cloud Deployment Scenarios from [4] for NR Standalone (SA) operation in Figures D-1 to D-7 and NR Non-Standalone (NSA) operation in Figures D-8 to D-11. Refer to [4] for up-to-date information and details on the Cloud Deployment Scenarios.

The O-CU and vO-CU referred in [4] should be considered as O-CU CP/UP and vO-CU CP/UP, respectively.

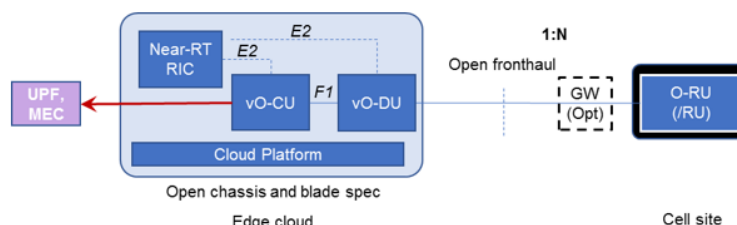


Figure D-1: Cloud Deployment Scenario A for SA operation [4]

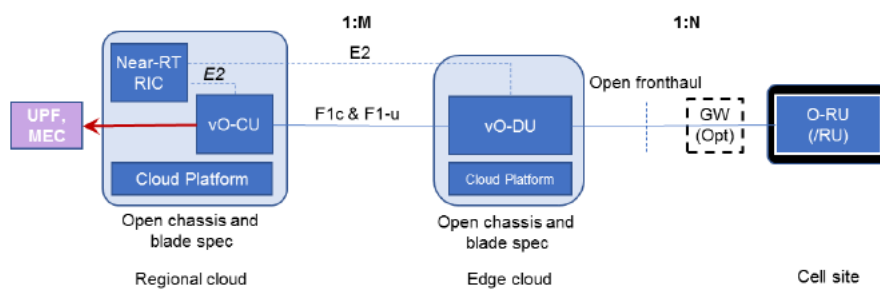


Figure D-2: Cloud Deployment Scenario C for SA operation [4]

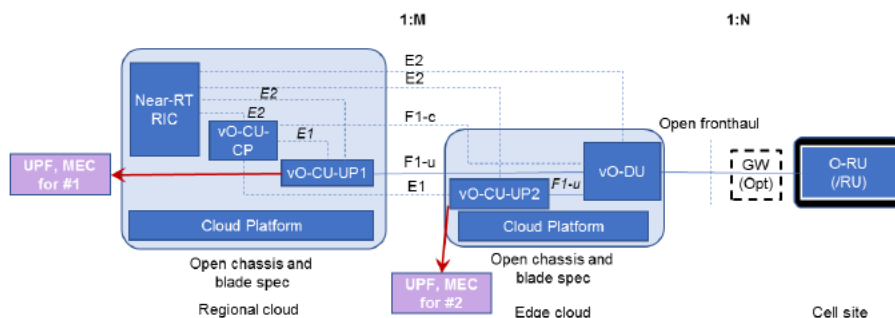


Figure D-3: Cloud Deployment Scenario C.1 for SA operation [4]

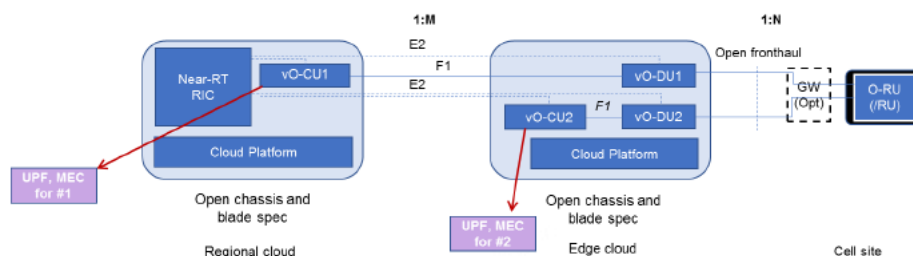


Figure D-4: Cloud Deployment Scenario C.2 for SA operation [4]

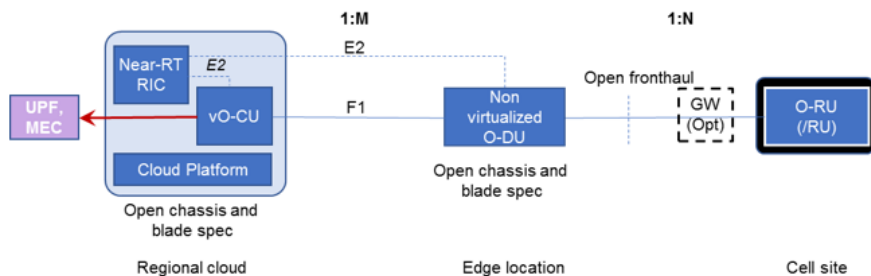


Figure D-5: Cloud Deployment Scenario D for SA operation [4]

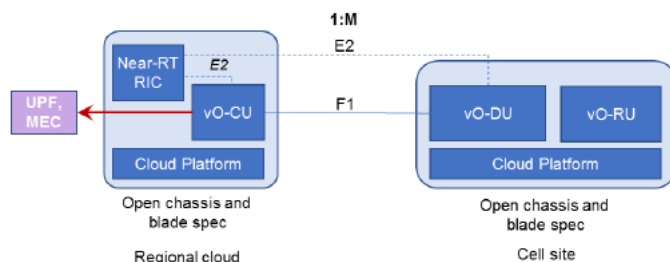


Figure D-6: Cloud Deployment Scenario E for SA operation [4]

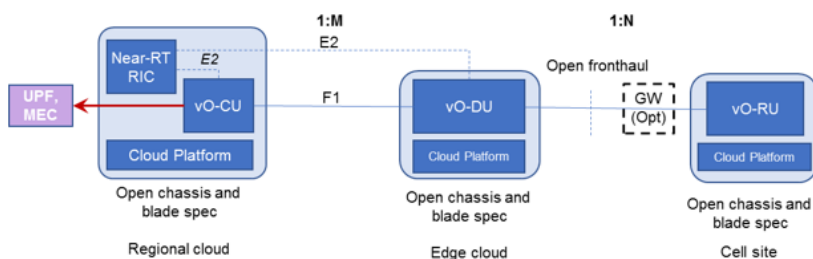


Figure D-7: Cloud Deployment Scenario F for SA operation [4]

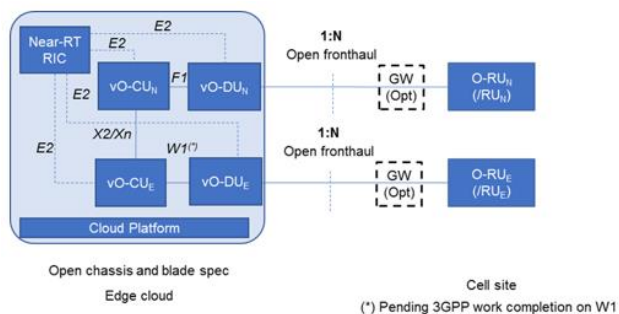


Figure D-8: Cloud Deployment Scenario A for NSA operation [4]

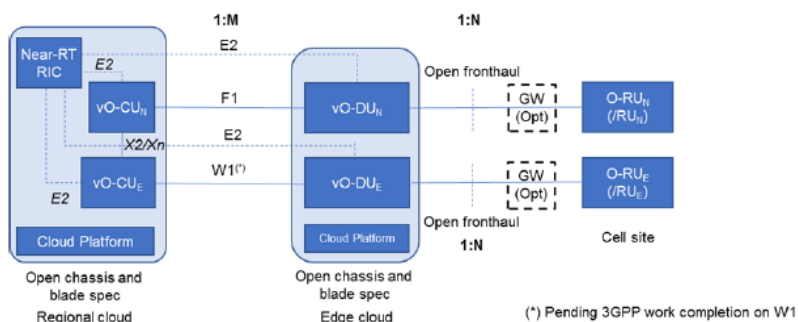


Figure D-9: Cloud Deployment Scenario C for NSA operation [4]

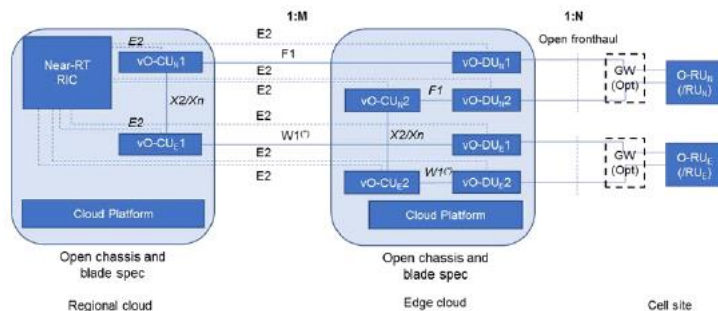


Figure D-10: Cloud Deployment Scenario C.2 for NSA operation [4]

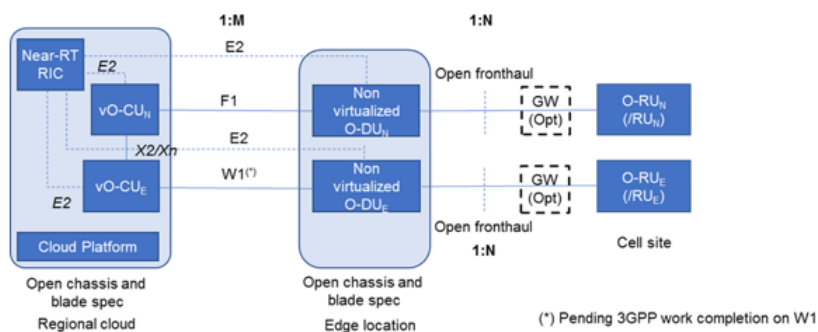


Figure D-11: Cloud Deployment Scenario D for NSA operation [4]

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